

Option-Implied Event Risk*

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Abstract

We introduce a nonparametric method for recovering risk distributions of anticipated events of publicly traded firms, such as earnings announcements, securities litigation events, and the resolution of takeover bids and proxy fights. We first apply our method to study three specific events: two corporate events and a systematic event (Elon Musk’s takeover bid of Twitter, the AT&T–Time Warner antitrust case, and the 2016 Brexit referendum). In all three cases, our recovered risk distributions are informative about market-implied beliefs over the outcome of the event. We then study a large sample of earnings announcements. Empirically, the recovered earnings announcement risk distribution is a strong predictor of realized announcement-day return moments, outperforming conventional option-implied measures that conflate announcement risk and other risks. Our method provides new insight into the effect of anticipated event risk on asset prices and how to measure such risk, and — more specifically — gives a fruitful way of using option prices to extract market beliefs about event risk.

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1 Introduction

In this paper, we introduce a parsimonious nonparametric framework to determine the information about event risk distributions contained in option prices over multiple expiration dates. We denote this the *event risk distribution recovery problem*. In many situations, the timing of an event that will affect an asset’s price is anticipated, but the effect it will have is not. An earnings announcement is one such example. Other examples include the verdict of a lawsuit, the timing of which may be known in advance, a planned shareholder vote, political uncertainty that is resolved by an election (see Kelly et al., 2016), or an FOMC announcement (see Lucca and Moench, 2015).

It is natural to use option markets to draw inferences about event risk. Indeed, the presence of instantaneous information events that discontinuously change prices provides an important explanation for why observed option prices deviate from those predicted by standard models, as manifested for example in implied volatility smiles and smirks across option strike prices, see the model in Black and Scholes (1973) and its many extensions.¹ Aït-Sahalia and Jacod (2009) find support for the presence of such instantaneous jumps in the market (see also Aït-Sahalia, 2004). Thus, the degree to which option prices deviate from what is predicted by standard models should be informative about the market’s view about future event risk.

A method to infer the market’s view about the event risk distribution would be helpful for forecasters and potentially for policy makers. It would also have practical value for traders, who face well-known challenges in calibrating implied volatility curves around earnings announcements. Indeed, a fundamental role of the market is to provide information signals, by aggregating and revealing information held by its participants, as argued by Hayek (1945). An important question is then how to disentangle the different pieces of information that together determine observed prices.

Our approach provides a way to disentangle an anticipated information event from the “normal” information diffusion that occurs regularly and continuously in the market. Our key, and almost only, assumption is that the normal information diffusion process is similar from one time-period to another. By studying options over multiple expiration dates, i.e., by using the term structure of option prices, we show that the market’s anticipation about the additional information during the announcement period can be recovered, as well as the distribution of normal information diffusion. This idea is very simple, but several theoretical and practical challenges are associated with the development of an operational method under weak assumptions and without imposing a particular parametric structure—challenges that we address in the paper.

Our method is based on convolution operators in infinite-dimensional discrete state spaces,

¹A large number of extensions and variations of the original Black-Scholes option pricing model have been developed. Early extensions include implied tree and local volatility models (see Rubinstein, 1994; Dupire, 1994; Derman and Kani, 1994), constant elasticity of variance (CEV) models (Cox, 1975), stochastic volatility models (Heston, 1993), models that incorporate isolated jumps (see Merton, 1976; Kou, 2002; Carr and Wu, 2003), the variance gamma model (Madan et al., 1998), and more generally time-changed Lévy processes (see Carr et al., 2002).

allowing us to use powerful results from transform theory (specifically, the theory of Z-transforms and Fourier analysis). We also address challenges of robustness associated with noisy observations of option prices, by introducing smoothing and convexification filters for the data. Technical details are provided in the Appendix, whereas the treatment in the body of the paper gives a less formal description of the method.

An important distinction is whether event risk is idiosyncratic or systematic. An earnings announcement for a small or medium-sized company would most likely have little or no impact on the market (see Leung and Santoli, 2014), whereas political uncertainty would likely have aggregate impact. In general, our model recovers risk neutral event distributions. When event risk is idiosyncratic, unless there are strong frictions, risk neutral and physical probabilities coincide and the interpretation of the results becomes particularly simple.

In the empirical section, we study several specific examples of anticipated event risk. We first analyze two illustrative examples that are well-suited for our approach, namely Elon Musk’s takeover bid of Twitter, and the AT&T–Time Warner antitrust case. We also study the 2016 UK Brexit vote, an event that hardly qualified as idiosyncratic. Nevertheless, our recovered risk-neutral event distribution is informative and matches implied probabilities observed in betting markets at that time quite well. Our option-implied probability of a Leave vote also begins to rise several days before the VIX does, and it tracks the betting odds far more closely.

We next focus on event risk associated with earnings announcements. An advantage of studying such events is data availability. Beyond recovering the announcement distribution, our decomposition also yields the normal-period distribution: the risk-neutral density for the week following the announcement. We validate this recovery empirically by comparing the recovered normal-period density to the ex-post risk-neutral distribution observed from options traded after the announcement, and show that it outperforms naive alternatives that do not separate announcement from normal-period risk.

Our paper relates to multiple strands of the literature. An extensive strand studies option prices around earnings announcements (see, e.g., Ravagli et al., 2014; Donders et al., 2000; Patell and Wolfson, 1981; Truong et al., 2012; Anagnostopoulou and Tsekrekos, 2015, 2017; Chung and Louis, 2017; Diavatopoulos et al., 2012; Ederington and Lee, 1996; Isakov and Pérignon, 2001; Jiang et al., 2012; Lei et al., 2020; Dubinsky et al., 2019; Alexiou et al., 2025; Liu et al., 2025). Our distinct contribution is our focus on recovering the entire event risk distribution from observed option prices. Closest to our approach in this strand is Alexiou et al. (2025), who document that short-maturity implied volatility curves often turn concave ahead of earnings announcements, and who read this as a sign of a bimodal risk-neutral distribution for the stock price. Our method recovers that distribution itself, and separates it from the normal-period risk that is also embedded in option prices.

A direct application of our method in this setting is to quantify the asymmetric tail risk sur-

rounding earnings announcements and to relate that risk to how option markets price announcement-related skewness and downside exposure. This is closely related to recent evidence that skew premia around earnings announcements are economically meaningful in equity options markets (Neururer and Papadakis, 2026). More broadly, firm-level differences in announcement tail risk may also be related to mechanisms emphasized in the literatures on disagreement, short-sale constraints, and overvaluation (Boehme et al., 2006; Diether et al., 2002).

A second application is to connect the implied event distribution to the evidence from option markets that higher risk-neutral moments measured before earnings announcements contain information about subsequent returns. In particular, Diavatopoulos et al. (2012) show that changes in option-implied skewness and kurtosis prior to earnings announcements predict subsequent stock and option returns, while Xing et al. (2010) show more generally that the shape of the individual-option volatility smirk predicts future equity returns and is linked to subsequent earnings shocks.

A related strand of the literature uses the maturity structure of option prices to learn about future information events, exploiting the fact that an anticipated event affects options differently depending on the horizon over which they are exposed to it. Patell and Wolfson (1981) show that implied volatilities rise ahead of quarterly earnings announcements and fall once the announcement has passed. Dubinsky et al. (2019) show that a scheduled announcement makes the term structure of implied volatility slope downward, since the announcement-day jump is spread over a shorter horizon in shorter-dated options, and use the implied volatilities of two options expiring after the announcement to estimate the volatility of the jump. Their estimator assumes that the log price jump is normally distributed, so that prices remain lognormal and the term structure identifies only the jump’s volatility, not its shape. In the context of mergers, Brown et al. (1994) show that in a sample of cash tender offers, the implied volatilities of target-firm options expiring before and after the deal’s resolution date follow a pattern consistent with prices anticipating the eventual outcome, which lets them infer both the probability and the timing of success. Bester et al. (2023) develop a model of target option prices in cash mergers in which implied volatility approaches zero as the probability of deal success approaches one, allowing them to estimate that probability from option prices. Similarly, Kelly et al. (2016) find that options whose lives span national elections and global summits are more expensive than otherwise similar options that do not. These papers use the pattern of prices across maturities to measure a single summary of the event, such as its variance or the probability that it occurs. Our approach instead uses options expiring on both sides of the event to recover the entire distribution of the event-related price change, separately from the distribution of normal-period risk.

Another related strand studies the general recovery of objective probability distributions and pricing kernels from observed state prices (see Ross, 2015; Carr and Yu, 2012; Backwell, 2015; Jensen et al., 2019; Qin and Linetsky, 2016; Borovička et al., 2016; Walden, 2017; Schneider and Trojani, 2019; von Sydow and Walden, 2020). Our method has similarities with this strand, but

our environment is somewhat different. In particular, we do not rely on time-homogeneous state transitions, since a key characteristic of the announcement is that it is different from normal innovations.

Finally, our paper is related to the literature on jumps in asset prices (see Ait-Sahalia, 2004; Busch et al., 2011; Lee, 2012; Seo and Wachter, 2019; Duong and Swanson, 2015; Jiang and Tian, 2010; Pan, 2002; Eraker et al., 2003; Eraker, 2004; Subramanian, 2004). Our approach differs from this literature by focusing on anticipated events, and by using option prices to draw inferences about the distribution of specific future jumps, under weak assumptions.

The rest of the paper is structured as follows. In Section 2, we introduce the model. In Section 3, we study several event risk environments — specifically some firm-specific events and a systematic event, as well as a larger set of earnings announcements — all of which have in common that the timing of the event is *ex ante* known but its impact on asset prices is not. We document that the recovered event risk distribution is informative about actual event risk in all of these environments, and also that the recovered normal-period distribution predicts the *ex-post* risk-neutral distribution observed after the announcement. Section 4 concludes. Proofs and additional examples are provided in the appendix.

2 Model

We describe our approach in a simple setting, since much of the intuition of the model can be captured in such a setting with three information event dates and a small number of states. We then provide an overview of the full model, which is described and analyzed in detail in Appendix A.

2.1 A simple setting

The time right now is 0. An announcement is scheduled at $t = 0.1$, which may affect the stock price of a publicly traded company. Specifically, there are five possible announcement outcomes, represented by the variable a . We use the term announcement for this event—an example would be an earnings announcement—but there are many other events that our framework can be applied to, e.g., a shareholder vote. The news may be “bad” ($a = -2$), “quite bad” ($a = -1$), “neutral” ($a = 0$), “quite good” ($a = 1$), or “good” ($a = 2$), and the probabilities for these outcomes are p_a , $a \in \{-2, -1, 0, 1, 2\}$. The associated probability vector is $p = (p_{-2}, p_{-1}, p_0, p_1, p_2)$. The announcement news is firm-specific and therefore not associated with a risk premium, so p also corresponds to the risk neutral probability distribution of the announcement outcomes. Our goal is to recover the probability distribution from option prices. We call this the *event risk distribution recovery problem*.

We may think of a period as being relatively short, e.g., a month, a week, or even a day. During a period, “normal” news arrives gradually. Normal news during a period has seven pos-

sible outcomes, $m \in \{-3, -2, -1, 0, 1, 2, 3\}$, with associated risk-neutral probability vector $q = (q_{-3}, q_{-2}, q_{-1}, q_0, q_1, q_2, q_3)$. These risk-neutral probabilities are time-invariant, i.e., they have the same distribution at $t = 2$ (with realization m^2) as at $t = 1$ (with realization m^1). The announcement event at $t = 0.1$ is embedded in the normal diffusion of information that occurs between $t = 0$ and $t = 1$.

The effect on the stock price of the cumulative news until $t \in \{1, 2\}$ is captured by the state variable $s^t \in \Omega = \{-8, -7, \dots, -1, 0, 1, \dots, 7, 8\}$, with the size of the state space $S = |\Omega| = 17$. Specifically, the stock price $P^t = G(s^t)$ for some increasing function $G : \Omega \rightarrow \mathbb{R}_+$. The state at time 1 is then $s^1 = a + m^1$, and the state at time 2 is $s^2 = a + m^1 + m^2$. A useful example is when G is a linear function, $G(s) = \alpha + \beta s$, $\beta > 0$.

Options on the company's stock with expirations at $t = 1$ and $t = 2$ are traded in the market. Their prices can be used to infer state-contingent Arrow-Debreu (AD) security prices, i.e., the time-0 prices, Ψ_s^t , of contingent claims that pay a dollar in state s at $t \in \{1, 2\}$. The gross risk free rate is assumed to be constant, $R > 0$. It follows that the risk-neutral probability of being in state s at $t \in \{1, 2\}$ is $q_s^t = R^t \Psi_s^t$. These risk-neutral probabilities are summarized in the probability vectors q^t , $t \in \{1, 2\}$, and are observable for $t \in \{1, 2\}$ (but *not* for $t = 0.1$). This pivotal assumption captures the fact that there are usually no claims traded in the market contingent on the outcome of specific announcement events.

The information environment can be described by a state transition tree, as shown in Figure 1. The tree has five branches between time 0 and 0.1, representing the five possible outcomes of the news announcement, and then seven branches in each of the periods between 0.1 and 1, and between 1 and 2. We note that it is straightforward to generalize the model to longer horizons than $T = 2$ and/or to environments with more possible news outcomes, represented by longer probability vectors, p and q .

If we observe q^1 and q^2 , we should in principle have sufficient information to recover both p and q . Indeed, we have a system of $2+11+17=30$ nonlinear equations (from total probabilities summing to unity, q^1 , and q^2) that we can use to infer the values of the 12 unknowns (5 p 's and 7 q 's). This should suffice. One may even conjecture that q^2 may be disregarded, and only the $2+11=13$ remaining equations are needed, since there are only 12 unknowns. This intuition is incorrect, and both q^1 and q^2 are needed in general.

We provide a simple example with only three possible outcomes for the announcement and normal news events: Specifically,

$$p = \left(0, \frac{1}{7}, \frac{4}{7}, \frac{2}{7}, 0\right), \quad \text{and} \quad q = \left(0, 0, \frac{1}{4}, \frac{1}{2}, \frac{1}{4}, 0, 0\right). \quad (1)$$

This corresponds to an economy with a trinomial tree for news, as seen in Figure 2. Note that in this case, only 5 states are reachable by $t = 1$: $s \in \{-2, -1, 0, 1, 2\}$, and only 7 by $t = 2$:

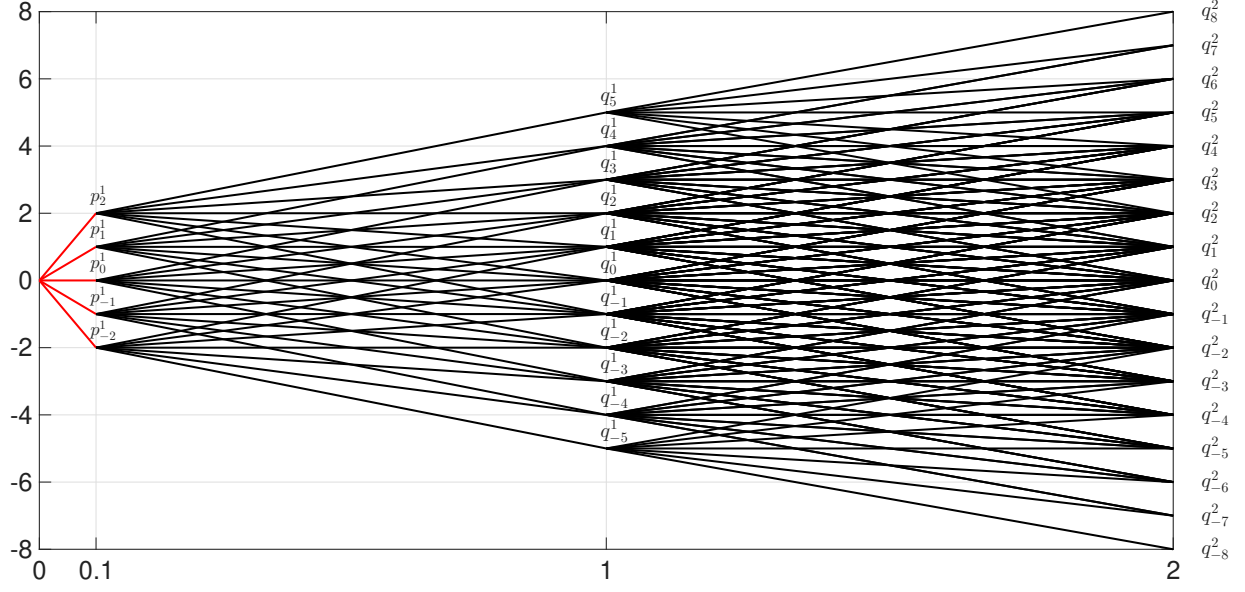


Figure 1: *State transition tree.*

$s \in \{-3, -2, -1, 0, 1, 2, 3\}$.

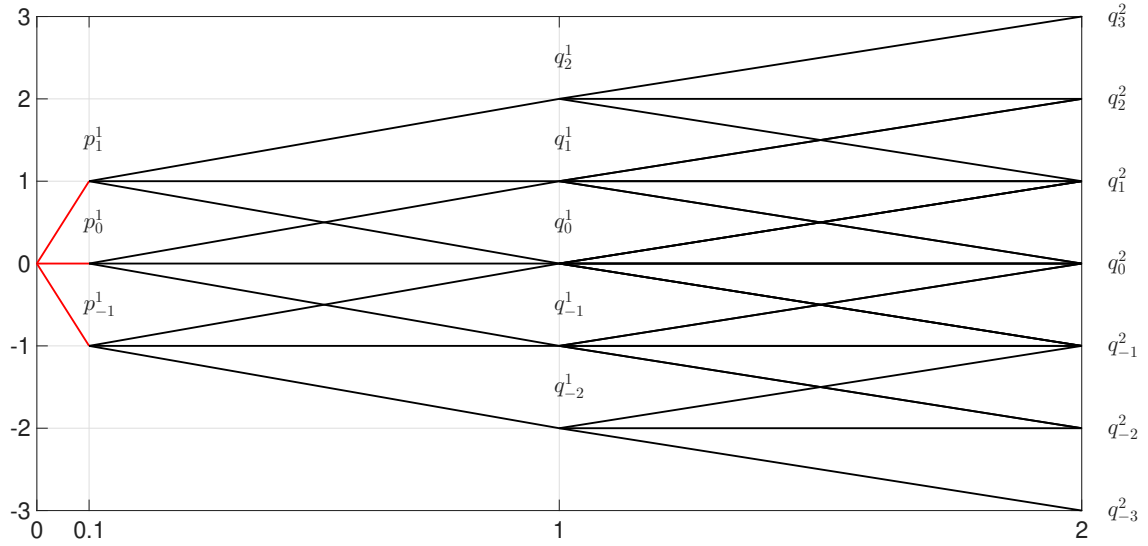


Figure 2: *State transitions in simplified example with a trinomial tree.*

There are 6 unknown probabilities here (p_{-1} , p_0 , p_1 , q_{-1} , q_0 , and q_1), and from the figure it

follows that the time-1 state prices, q^1 , can be calculated from the following equations,

$$\begin{aligned} q_{-2}^1 &= p_{-1}q_{-1}, \\ q_{-1}^1 &= p_{-1}q_0 + p_0q_{-1}, \\ q_0^1 &= p_{-1}q_1 + p_0q_0 + p_1q_{-1}, \\ q_1^1 &= p_0q_1 + p_1q_0, \\ q_2^1 &= p_1q_1, \end{aligned}$$

leading to

$$q^1 = \left(0, \frac{1}{28}, \frac{3}{14}, \frac{11}{28}, \frac{2}{7}, \frac{1}{14}, 0\right)'. \quad (2)$$

Additionally,

$$p_{-1} + p_0 + p_1 = 1, \quad \text{and} \quad q_{-1} + q_0 + q_1 = 1,$$

so altogether we have 7 equations for 6 unknowns. The correct probability vectors given in (1) solve this system of equations, but one verifies that in addition the incorrect probability vectors

$$p = \left(0, \frac{1}{14}(3 - \sqrt{2}), \frac{1}{2}, \frac{1}{14}(4 + \sqrt{2}), 0\right), \quad q = \left(0, 0, \frac{1}{14}(3 + \sqrt{2}), \frac{1}{2}, \frac{1}{14}(4 - \sqrt{2}), 0, 0\right)$$

also provide a solution, so uniqueness fails. There are also two additional solutions, obtained by switching p and q for both the correct and incorrect solutions above.

This issue of multiple solutions when only using q^1 grows exponentially with the sizes of the p and q vectors. We therefore focus on methods that use options with two maturity dates, both q^1 and q^2 in this example. The approach captures the idea that important information about the risk distribution for individual events can be recovered from the term structure of option prices, rather than focusing solely on the cross section of prices on a single expiration date. When both q^1 and q^2 are used, the correct solution is in general unique, i.e., p and q are uniquely identified. However, solving the system of nonlinear equations becomes challenging as the number of elements in the p and q vectors and/or the time horizon increases.

2.2 The general model

We generalize the previous setting to a model that: (i) works for arbitrarily long probability vectors, p and q ; (ii) works for longer time horizons, $T > 2$; and (iii) is numerically fast and stable. Our theoretical analysis is based on the theory of Z -transforms, and the numerical method is based on the Fast Fourier Transform (FFT). The model is explained in detail in Appendix A, whereas we here provide an overview, building on the previously introduced simple example.

As a link to the general model, we show how the simple example can be defined in terms of

operators. In this finite-dimensional setting, we can define the 7×7 matrices

$$A = \begin{bmatrix} \frac{4}{7} & \frac{1}{7} & 0 & 0 & 0 & 0 & 0 \\ \frac{2}{7} & \frac{4}{7} & \frac{1}{7} & 0 & 0 & 0 & 0 \\ 0 & \frac{2}{7} & \frac{4}{7} & \frac{1}{7} & 0 & 0 & 0 \\ 0 & 0 & \frac{2}{7} & \frac{4}{7} & \frac{1}{7} & 0 & 0 \\ 0 & 0 & 0 & \frac{2}{7} & \frac{4}{7} & \frac{1}{7} & 0 \\ 0 & 0 & 0 & 0 & \frac{2}{7} & \frac{4}{7} & \frac{1}{7} \\ 0 & 0 & 0 & 0 & 0 & \frac{2}{7} & \frac{4}{7} \end{bmatrix}, \quad \text{and} \quad N = \begin{bmatrix} \frac{1}{2} & \frac{1}{4} & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{4} & \frac{1}{2} & \frac{1}{4} & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{4} & \frac{1}{2} & \frac{1}{4} & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{4} & \frac{1}{2} & \frac{1}{4} & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{4} & \frac{1}{2} \end{bmatrix}, \quad (3)$$

and the vector $\delta = (0, 0, 0, 1, 0, 0, 0)'$. We then have the matrix-vector relations

$$(q_{-3}^1, q_{-2}^1, q_{-1}^1, q_0^1, q_1^1, q_2^1, q_3^1)' = NA\delta, \quad \text{and} \quad (q_{-3}^2, q_{-2}^2, q_{-1}^2, q_0^2, q_1^2, q_2^2, q_3^2)' = N^2A\delta. \quad (4)$$

The event risk distribution recovery problem for this simple example can then be stated in terms of recovering the matrix A from observing q^1 and q^2 . Note that this is not a standard linear matrix inversion problem, since it is the matrix A that is unknown.

This formulation can now be extended to a general setting, in which there are T periods, state prices are observed at times $t < T$ and T , with $t, T \in \mathbb{N}$ (where $\mathbb{N}$ denotes the set of natural numbers), and the announcement event occurs at time $u \in (t^*, t^* + 1)$, where $t^* \in \{0, 1, \dots, T-1\}$. The sequencing of events is shown in Figure 3.

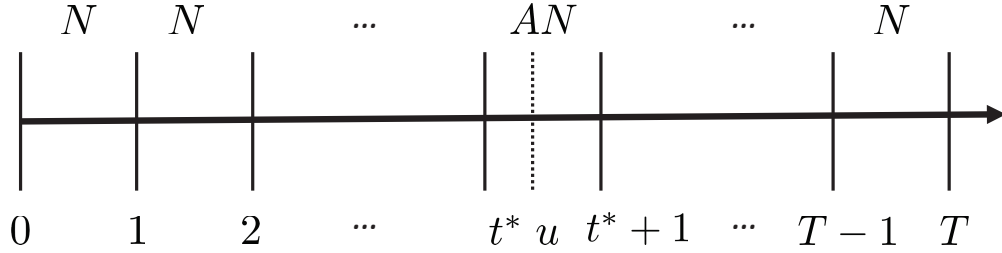


Figure 3: Sequence of normal (N) and announcement (A) information events. Announcement occurs at $u \in (t^*, t^* + 1)$, where $t^* \in \{0, 1, \dots, T-1\}$.

There are two cases. The announcement may occur before the first observation, $u < t$, in which case

$$q^t = N^{t-t^*} AN^{t^*} \delta, \quad \text{and} \quad q^T = N^{T-t^*} AN^{t^*} \delta. \quad (5)$$

Alternatively, it may occur between t and T , $t < u < T$, in which case

$$q^t = N^{t^*} \delta, \quad \text{and} \quad q^T = N^{T-t^*} AN^{t^*} \delta. \quad (6)$$

Also, in the general setting the announcement and normal probability vectors may be arbitrarily long, $p = (p_{-K_p}, p_{-K_p+1}, \dots, p_0, \dots, p_{K_p-1}, p_{K_p})$, and $q = (q_{-K_q}, q_{-K_q+1}, \dots, q_0, \dots, q_{K_q-1}, q_{K_q})$, $K_p \in \mathbb{N}$, $K_q \in \mathbb{N}$. There will then be $S = 2(K_q T + K_p) + 1$ reachable states, the matrices corresponding to (3) will be $S \times S$ matrices, and the corresponding matrix equations in (5,6) will represent systems of high-order multivariate algebraic equations (up to order $T + 1$). In the simple example, $K_q = 1$, $K_p = 1$, and $T = 2$.

We note that since K_q , K_p , and T may be arbitrarily large, rich dynamics can be captured by the general model. For example, a period may capture the weekly returns of the underlying asset, with options available that expire $t = 9$ and $T = 11$ weeks into the future, and the announcement occurring 12 days ahead. The number of possible announcement outcomes may be 15 ($K_p = 7$), and the number of normal period outcomes 21 ($K_q = 10$), leading to in total $S = 2(10 \times 11 + 7) + 1 = 235$ reachable states, with no parametric restrictions imposed on the probability vectors p and q . Inasmuch as discrete processes can approximate continuous processes arbitrarily well, our nonparametric discrete model has few restrictions.

The key assumption we impose is that normal transitions are time homogeneous, i.e., that normal period events have the same risk neutral distribution in each period. This assumption is not innocent. If one believes next week's normal return distributions will be significantly different from this week's—beyond the difference because of the announcement event—then our model is misspecified. One may conjecture that the assumption is more appropriate over fairly short periods—during which there is little time for the economic environment to fundamentally change—as well as over long time periods—during which temporary deviations from longer-term dynamics are less important. Under intermediate horizons, the assumption may be less appropriate, but it is ultimately an empirical question how useful our method is in practice.

Our second significant assumption is that the announcement event is idiosyncratic, so that its risk-neutral and objective probabilities coincide. If this is not the case, our method recovers the risk-neutral probability distribution. Again, it is an empirical question whether the recovered distribution is informative in that case.

Mathematically, we represent news events as so-called Laurent operators on the infinite state space, $\Omega = \mathbb{Z}$ (where $\mathbb{Z}$ is the set of integers). These operators are a type of convolution operators, as a consequence of our assumption of time-homogeneity. A representative agent has expected utility preferences over aggregate consumption in these states, such that the corresponding equilibrium stochastic discount factor has a risk-premium associated with normal news events, but not with announcement events. Risk-neutral and objective probabilities therefore coincide over the announcement part of the state space. Allowing the state space to be infinite allows a unified treatment for arbitrarily large parameter values, K_p , K_q , t , and T . Intuitively, the finite-dimensional matrices in (3), with dimensionalities that depend on parameter values, are replaced by bi-infinite matrices in this setting.

2.3 The general recovery result and the practical method

The key to overcoming the discussed challenges when recovering p is that the involved operators are convolution operators, allowing us to use the theory of Z -transforms and Fourier analysis to solve the problem, see Katznelson (1976) and Jackson (1989). Formally, the Z -transform of a probability vector is a function defined as

$$P(z) = \mathcal{Z}[p](z) = \sum_k p_k z^{-k},$$

defined over the complex numbers, $z \in \mathbb{C} \setminus \{0\}$. When p has a finite number of nonzero elements, the function P is a so-called Laurent polynomial. The following theoretical result is derived in Appendix A:

Theorem 1 (Theoretical recovery). *Assume $T = rx$, where $r, x \in \mathbb{N}$, $r < T$. From the observed risk-neutral probabilities q^r , q^{T-r} , and q^T , define the Laurent polynomials*

$$X(z) = \sum_k q_k^r z^{-k}, \quad V(z) = \sum_k q_k^{T-r} z^{-k}, \quad W(z) = \sum_k q_k^T z^{-k}.$$

(i) *If $u > r$, then the announcement distribution can be recovered as*

$$p_k = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{ik\omega} \frac{W(e^{i\omega})}{X(e^{i\omega})^x} d\omega. \quad (7)$$

(ii) *If $u < T - r$, then the announcement distribution can be recovered as*

$$p_k = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{ik\omega} \frac{V(e^{i\omega})^x}{W(e^{i\omega})^{x-1}} d\omega. \quad (8)$$

The theorem shows that the event distribution can be uniquely identified from state-prices at two dates, with different formulas depending on whether the announcement occurs before or after the first date. We note that the theorem puts some restrictions on t and T , namely that $T = tx$ if $u > t$, and $T = t(x+1)$ if $u < t$, where x is a whole number.

Two practical challenges remain: first, how to evaluate the integral, either (7) or (8) in the theorem; and second how to calculate state prices from option prices. For the first challenge, we use the fast Fourier transform (FFT) to represent the Laurent polynomials, and its inverse (IFFT) for an efficient approximation of the integral. For probabilities q with $q_k = 0$ when $|k| > K$, the FFT is defined by

$$\hat{q}_\omega = \mathcal{F}[q](\omega) = \sum_{k=-K}^K q_k e^{-2\pi i k \omega / (2K+1)}, \quad \omega = -K, -K+1, \dots, K, \quad (9)$$

and the IFFT by

$$q_k = \mathcal{F}^{-1}[\hat{q}](k) = \frac{1}{2K+1} \sum_{\omega=-K}^K \hat{q}_\omega e^{2\pi i k \omega / (2K+1)}, \quad k = -K, -K+1, \dots, K. \quad (10)$$

Further details are provided in Appendix B. The parameter K here should be chosen sufficiently large to cover all feasible states up until time T , i.e., $K \geq K_q T + K_p$ in the model. In practice, K should be chosen sufficiently large to ensure that time- T state prices are effectively zero outside of the states $-K \leq s \leq K$. Henceforth we use the notation $q \in \ell_K^0$, if $q_k = 0$ when $|k| > K$, and this condition is then that K should be chosen such that $q^T \in \ell_K^0$.

For the second challenge, we may infer state prices from option prices using standard differencing methods, see Breeden and Litzenberger (1978). For us, however, it is preferable to work directly with option prices rather than state prices, because repeated differencing tends to amplify noise, and the transforms described in Theorem 1 are sensitive to such noise. We therefore derive a direct FFT-based recovery method from option prices. We here focus on the case $T = 2$ and $0 < u < 1$, although the method can be generalized to arbitrary u , t and T .

Consider time- t risk-neutral probabilities q^t , and an underlying asset with linear payoff, such that prices are

$$S_k = G(s_k) = \begin{cases} k + K, & k \geq -K, \\ 0, & k < -K. \end{cases} \quad (11)$$

A put option with maturity t and strike S_r , $r \geq -K$, has price

$$P_r = \begin{cases} 0, & r = -K, \\ \sum_{k=-K}^{r-1} (r-k) q_k^t, & r = -K+1, -K+2, \dots, K, \\ P_K + (r-K), & r > K. \end{cases} \quad (12)$$

Hence

$$q_k^t = \begin{cases} P_{k+1}, & k = -K, \\ P_{k-1} - 2P_k + P_{k+1}, & k = -K+1, -K+2, \dots, K, \end{cases}$$

that is, risk-neutral probabilities are obtained from put prices by second differencing.²

We also define the $(2K+1)$ -periodic second-difference operator $\widehat{D_+ D_-}$, which coincides with the usual second-order difference operator in the interior and is completed periodically at $-K$ and K :

$$\begin{aligned} \left(\widehat{D_+ D_-} P \right)_{-K} &= P_K - 2P_{-K} + P_{-K+1}, \\ \left(\widehat{D_+ D_-} P \right)_K &= P_{K-1} - 2P_K + P_{-K}. \end{aligned}$$

²So P contains the same information as the risk neutral probabilities, q^t , in line with the observation in Breeden and Litzenberger (1978).

Altogether, this leads to a method for practically recovering the announcement event distribution directly from put option prices, using the FFT and its inverse, as shown in the following theorem:

Theorem 2 (Practical recovery). *Consider the model with $T = 2$ and $0 < u < 1$, with put prices $P^1, P^2 \in \ell_K^0$. Define adjusted prices $\dot{P}^1, \dot{P}^2 \in \ell_K^0$ by*

$$\dot{P}_k^1 = P_k^1 - \frac{k+K}{2K+1} \left(P_K^1 - \frac{1}{2}(2K+2) \right) - \frac{1}{2} \frac{(k+K+1)(k+K+2)}{2K+1}, \quad (13)$$

$$\dot{P}_k^2 = P_k^2 - \frac{k+K}{2K+1} \left(P_K^2 - \frac{1}{2}(2K+2) \right) - \frac{1}{2} \frac{(k+K+1)(k+K+2)}{2K+1}, \quad (14)$$

for $k = -K, -K+1, \dots, K$. Define

$$\hat{P}^i(\omega) = \mathcal{F} \left[\dot{P}^i \right] (\omega), \quad i = 1, 2, \quad (15)$$

and

$$F \stackrel{\text{def}}{=} \mathcal{F}^{-1} \left[\frac{\hat{P}^1(\omega)^2}{\hat{P}^2(\omega)} \right]. \quad (16)$$

Then the announcement event distribution p satisfies

$$p_k = \begin{cases} \left(\widehat{D_+ D_-} F \right)_k + \frac{1}{2K+1}, & |k| \leq K, \\ 0, & |k| > K. \end{cases} \quad (17)$$

This direct option-price method has multiple advantages over first recovering state prices and then applying Theorem 1. Most importantly, it uses second differencing only once, in the final step, which reduces noise amplification. In addition, asynchronous or noisy option prices can generate negative implied state prices, in violation of basic no-arbitrage principles. This can be avoided by imposing convexity constraints on P^1 , P^2 , and the put-price schedule implied by p . We return to this issue in the empirical section. Finally, the method uses only FFT operations and is therefore computationally very fast.

2.4 A special case: two-point-distributed event risk

In some cases, one may have additional information about the event risk. For example, the event may represent a shareholder vote whether to approve an acquisition with a binary Yes/No outcome. In this case, it is known that p only has two nonzero values, $p_{k_1} > 0$ and $p_{k_2} > 0$, $k_1 < k_2$, although the sizes of the jumps (the values of k_1 and k_2) are not directly observable. Now, the price of the underlying asset right before the event equals its expected price right after the event (the event is idiosyncratic so there is no risk premium),³ $P^{u-} = E[P^u]$, which given (k_1, k_2) uniquely determines

³For a systematic event, such as the Brexit referendum in Section 3, the same relation holds with the expectation taken under the risk-neutral measure.

the probabilities (p_{k_1}, p_{k_2}) . Also, note that for (k_1, k_2) to be feasible, both probabilities must be positive, implying that $P^u(k_1) < P^{u-} < P^u(k_2)$.

This argument suggests an alternative method to recover the event risk distribution, namely by iterating over different combinations of feasible (k_1, k_2) , and for each realization checking whether the two-point distribution matches observed state prices, as formalized by the following theorem:

Theorem 3 (Two-point distribution). *From the observed risk-neutral probabilities q^t and q^T , when the announcement event has a two-point distribution determined by (k_1, k_2) , define the Laurent polynomials*

$$X(z) = \sum_k q_k^t z^{-k}, \quad W(z) = \sum_k q_k^T z^{-k}, \quad B_{\ell_1, \ell_2}(z) = p_{\ell_1} z^{-\ell_1} + p_{\ell_2} z^{-\ell_2},$$

and

$$R_{\ell_1, \ell_2}^1(z) = \frac{W(z)^t}{X(z)^T B_{\ell_1, \ell_2}(z)^t}, \quad R_{\ell_1, \ell_2}^2(z) = \frac{X(z)^T}{W(z)^t B_{\ell_1, \ell_2}(z)^{T-t}}. \quad (18)$$

Then,

- (i) when $u > t$, $R_{\ell_1, \ell_2}^1 \equiv 1$ if and only if $(\ell_1, \ell_2) = (k_1, k_2)$,
- (ii) when $u < t$, $R_{\ell_1, \ell_2}^2 \equiv 1$ if and only if $(\ell_1, \ell_2) = (k_1, k_2)$.

A straightforward approach to recovering p is now to iterate over feasible combinations of (ℓ_1, ℓ_2) , calculate the appropriate $R_{\ell_1, \ell_2}(z)$ function, and the error

$$E = \frac{1}{2\pi} \int_{-\pi}^{\pi} |R_{\ell_1, \ell_2}(e^{i\omega}) - 1|^2 d\omega, \quad (19)$$

and pick the distribution that minimizes E .⁴ In practice, the error is approximated by $\sum_{\omega} |\hat{R}_{\omega} - 1|^2$, where the FFT is used to calculate $\hat{R}_{\omega}$, see (9). An advantage of this approach is that there are no joint restrictions on T and t (as were needed in Theorem 1), making it more flexible.

2.5 Examples

Four-period model with bimodal event risk

We first consider a market with $K_p = K_q = 30$, and $T = 4$. The risk neutral probability distribution during a normal period, q , has a double exponential distribution, as shown in the left panel of Figure 4. The announcement risk distribution is shown in the right panel, and is the mixture of two exponential distributions; it is bimodal. It could, for example, reflect the anticipated outcome of a lawsuit, where the company may be held liable and forced to pay a large fine. In this case, the

⁴We note that Parseval's theorem from Fourier analysis provides a direct link between this error measure in the transformed domain and the error in the physical domain. Specifically, for a probability distribution, p , and its estimate p^e , with Z -transforms $P(z)$ and $P^e(z)$, $\sum_k |p_k - p_k^e|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |P(e^{i\omega}) - P^e(e^{i\omega})|^2 d\omega$.

company's future cash flows will decrease sharply. On the other hand, if the company is cleared of liability, cash flows will be unaffected. The company's current valuation incorporates the risk of a potential negative outcome, so if it is cleared of liability, its stock price will increase.

The announcement occurs at $u = 2.5$. Both methods in Theorem 1 are then feasible: method (i) with $r = 2$ and $x = 2$, and method (ii) with $r = 1$ and $x = 4$. Method (i) uses q^2 and q^4 to recover p , whereas method (ii) uses q^3 and q^4 .

Figure 5 shows the risk neutral probabilities associated with contingent claims maturing between 1 and 4 periods ahead, q^1 , q^2 , q^3 , and q^4 . The leftmost panel, q^1 , is the same as that in Figure 4, since $u > 1$, and therefore $q^1 = q$. The effect of the announcement risk is quite pronounced in the third panel, q^3 , which gives the first probabilities that include the announcement at $u = 2.5$. The smoothing effect of the normal event risk over three periods is sufficient for the risk neutral probabilities to no longer be bimodal, but the bimodal announcement operator A still has a major effect. At $T = 4$, further smoothing has occurred, and the effect of the announcement operator is less pronounced.

Both methods (i) and (ii) work very well for this example and recover the announcement risk almost perfectly, with $\max_k |\hat{p}_k - p_k| \approx 2 \times 10^{-11}$ in both cases. We refer to Appendix D for further details.

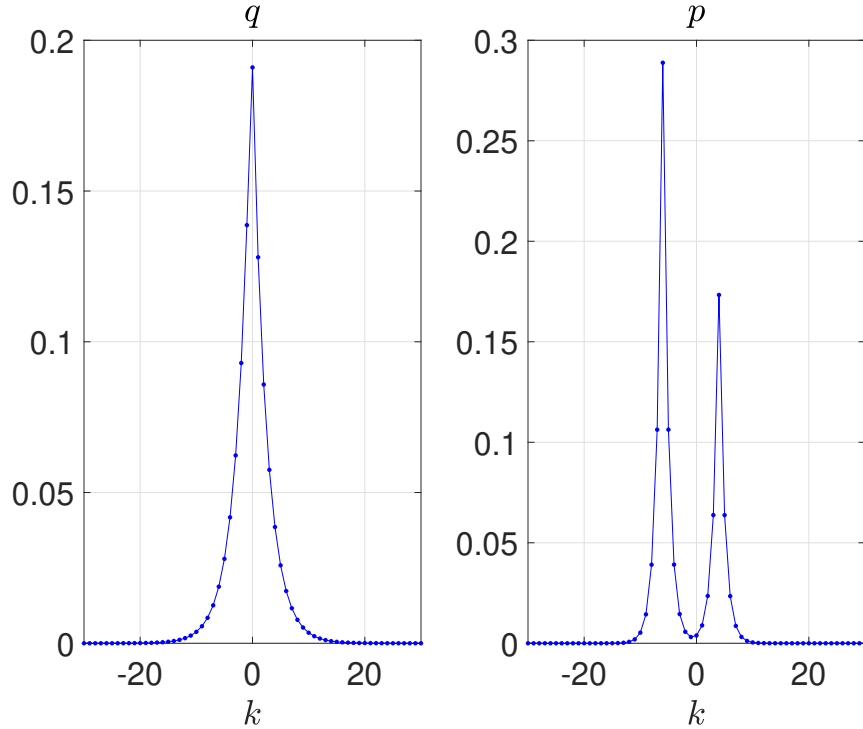


Figure 4: *Left: Risk neutral probabilities during normal period, q . Right: Announcement probabilities. Specification: $p_k = C_1 (e^{-|k+6|} + 0.6e^{-|k-4|})$, $q_k = C_2 e^{-0.4|k+0.1|}$.*

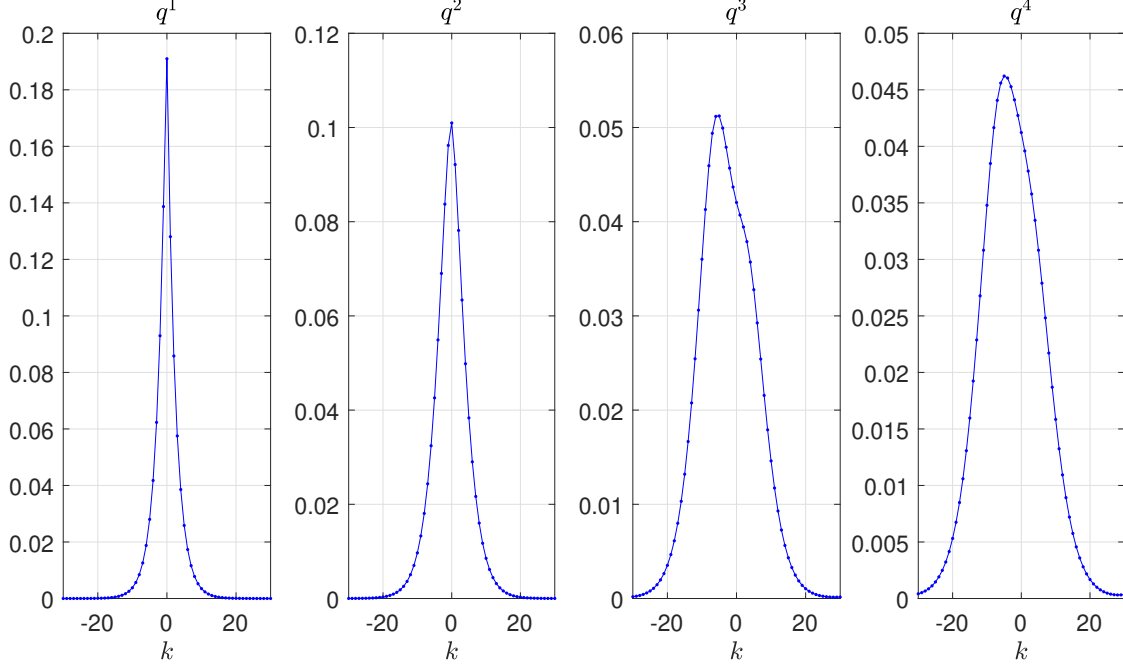


Figure 5: Risk neutral probabilities, q^1, q^2, q^3, q^4 .

Many-period binomial model with event risk

A second example is given by the binomial tree approximation of the Black-Scholes model with constant coefficients. Here $K_q = 1$, and the risk neutral probabilities during a normal period are $q_1 = \frac{R-\bar{d}}{\bar{u}-\bar{d}}$, $q_0 = 0$, and $q_{-1} = \frac{\bar{u}-R}{\bar{u}-\bar{d}}$, where $\bar{u}$ and $\bar{d} = 1/\bar{u}$ are the sizes of the up and down moves in the binomial tree. (We write the up move as $\bar{u}$ to keep it distinct from the announcement time u .) In this case, the state represents the value of a stock, $S_t = \bar{u}^k S_0$, in state k at time t .

To this we add announcement risk. We study an example with $T = 28$ time periods, each representing a day, and where the announcement occurs before $t = 14$. Thus, method (ii) is applicable, with $r = 14$ and $x = 2$. We choose $\bar{u} = 1/\bar{d} = e^{0.02}$, corresponding to a 2% continuously compounded volatility per day, i.e., 38.2% per year. The risk-free rate is normalized to $R = 1$.

Figure 6 shows four different examples, each with a different type of event risk. In row 1, the topmost row, there is no announcement, corresponding to $p_0 = 1$. The two leftmost panels show the risk neutral probabilities for 14 and 28 days into the future. These are basically binomially distributed, except that only probabilities for an even number of moves are non-zero, due to the recombining nature of the binomial tree. The next panel shows the 28-day implied (daily) volatilities associated with these risk neutral probabilities, across strike prices. We see that they are close to 0.02, the difference stemming from the approximation the binomial tree model provides of the Black-Scholes continuous time model. Finally, the rightmost panel shows the actual and recovered event risk, obtained from the risk neutral probabilities q^{14} and q^{28} using method (ii) in Theorem 1.

We see that the method basically perfectly identifies the event risk.

It is important to note that our method is nonparametric, in that it makes no assumptions about the form of normal and announcement risk distributions. Specifically, the fact that the normal-period dynamics are generated by binomial-tree dynamics is not used anywhere in the recovery method.

In row 2, a bell-shaped announcement risk distribution is used. This increases the spreads of the risk-neutral distributions (two leftmost panels), and increases the level of the implied volatility curve (third panel from the left) compared with the benchmark case. Qualitatively, these panels look as they would if the per-period volatility were higher than 0.02.⁵ Again, the event risk distribution is very well recovered.

In row 3, a bimodal announcement risk distribution is used. Specifically, the stock price either moves up by a quite specific amount ($k \mapsto k + 4$ or $k \mapsto k + 5$, with equal probability) represented by the sharp spike on the right, or moves down more diffusively by 1 to 11 steps, represented by the left hump. The effect on the implied volatility curve is to make it more “hump”-shaped on the left, because of the more narrowly defined downside compared with the wider bell-shaped risk shown in row 2, as well as flatter in the right part, because of the more limited potential upside move compared with row 2. As in the previous example, the recovered event risk distribution is very close to the actual risk distribution.

Finally, in row 4, a two-point announcement risk distribution is used. The corresponding implied volatility curve is now even further affected, having a “hat”-like shape. As in the previous cases, the recovered event risk distribution coincides closely with the actual one, as shown in the rightmost panel.

In Appendix D, we provide another example that corresponds to a trinomial tree version of the constant elasticity of variance (CEV) model with additional event risk. We show that this model is also efficiently handled by our event risk distribution recovery method.

Prediction of future normal risk distribution

In addition to predicting event risk distributions, our method can also be used to predict post-event normal-period risk distributions. This is useful, e.g., if one wants to understand what implied volatility will look like after an announcement event.

We compare our estimate of the future risk distribution (q) with alternative specifications that do not incorporate the fact that the announcement only affects dynamics during the announcement period. Consider a situation with $T = 2$ and $u = 0.5$, so that the announcement occurs in the first

⁵A couple of technical differences also arise. First, the risk-neutral distributions are now smooth, as seen in row 2. They are no longer zero for odd-numbered moves, since the announcement may shift the state both an even number and an odd number of steps. Second, the implied volatility curve is meaningfully defined over a wider range of strike prices, since the number of reachable states increases when K_p increases.

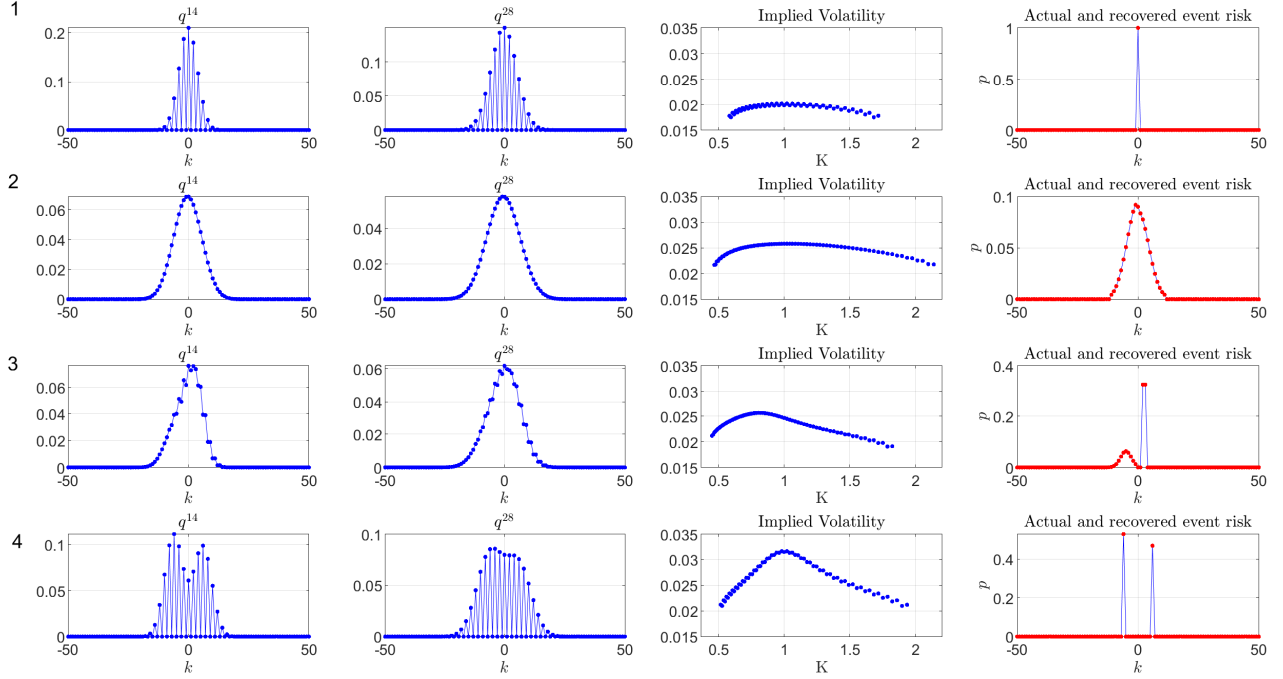


Figure 6: The figure shows risk neutral probabilities, implied volatilities and announcement risks in the Black-Scholes model with event risk. Row 1 panels have no event risk; Row 2 panels have bell-shaped event risk; Row 3 panels have bimodal event risk, combining a sharp upward spike with a diffuse downward hump; Row 4 has two-point distributed event risk. Column 1 shows risk neutral probabilities for $t = 14$; Column 2 shows risk neutral probabilities for $t = 28$; Column 3 shows the implied volatilities associated with call option prices; Column 4 shows actual (blue solid line) and recovered (red dotted line) event risk.

period. It follows that the operator in the first period is AN , and the operator in the second period is N (see Figure 3).

In the absence of an announcement, the second-period operator would be the same as the first-period operator (both would be N), and it could therefore be inferred from the first-period operator, or equivalently from the risk-neutral probabilities q^1 . The operator could also be inferred from the two-period risk-neutral probabilities, q^2 , by finding the operator N that leads to q^2 when applied twice. This can be thought of as taking the “square root” (technically in the Fourier domain) of q^2 . Both these estimates are incorrect in the presence of event risk.

In Figure 7, we provide a specific example. The upper panels show the normal (left) and announcement (right) event risk distributions. The lower panels show the estimated normal period risk distributions from q^1 (left) and q^2 (right). Both estimates significantly misrepresent the normal period risk distribution, q . Specifically, both methods to some extent include the announcement risk distribution, p , in the estimate. In contrast, our method almost perfectly recovers q in this example (the mean-squared error between estimated and actual q is 8×10^{-15} , due solely to double-precision rounding errors).

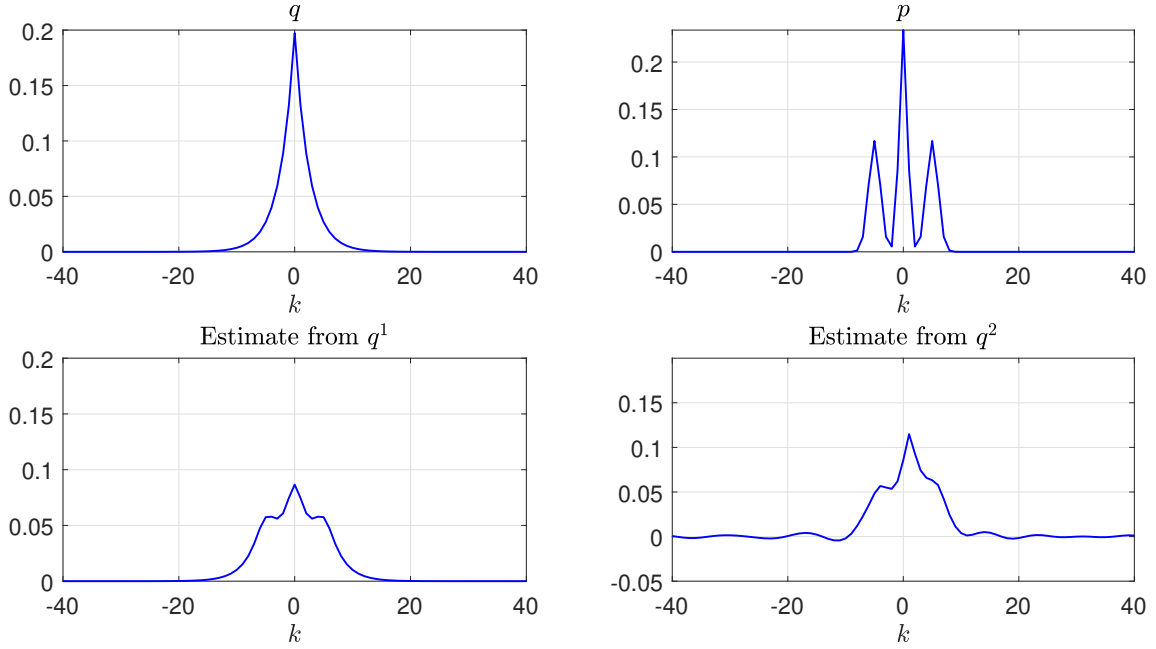


Figure 7: Upper left panel: Normal period risk distribution; Upper right panel: Event risk distribution; Lower left panel: Estimated normal period risk distribution from q^1 ; Lower right panel: Estimated normal period risk distribution from q^2 .

3 Empirical results

The empirical analysis asks whether option prices before an announcement contain useful information about the distribution of announcement-day returns. We first provide three empirical examples that illustrate the method: two high-profile firm-related announcements and one systematic event. We then describe our larger-scale analysis for firm-level earnings announcements, and we study how the recovered announcement distributions help predict realized returns. In both cases, the timing of the events is known *ex ante*, but the events’ impacts on share prices are not.

3.1 Three illustrative examples

To build intuition, we start with three examples to illustrate what the recovered announcement distributions look like in practice, before turning to the broader panel evidence. The first two are firm-level events, and the third is the 2016 Brexit referendum, an event with aggregate consequences.

A binary takeover announcement: Twitter and Elon Musk

Our first example considers the takeover of Twitter by Elon Musk in 2022. Musk agreed in April 2022 to acquire Twitter at \$54.20 per share, but he attempted to walk away from the agreement in May of that year after a share-price decline. Twitter responded by suing to enforce the merger. In early October, Musk signaled his willingness to complete the merger at the original price. The Delaware judge overseeing the case paused the lawsuit to provide time for the two parties to close the merger. The judge set an explicit deadline of October 28 for Musk to close the deal at the agreed-upon price, after which the case would go to trial.⁶ This decision created a clear, publicly known date by which the key announcement about the deal — whether the acquisition would close on the agreed terms — would be resolved. As of October 21, a week prior to the Friday, October 28 deadline, the deal had not closed. The week from October 21 to October 28 therefore provides an ideal setting for our approach: investors faced a single, well-defined announcement window, with little ambiguity about the timing of the news.

Figure 8 shows the recovered announcement distribution for Twitter based on options observed on Friday, October 21 and expiring on the following two Fridays (October 28 and November 4). The recovery procedure is based on Theorem 2, and we describe it in detail below. The distribution has a pronounced mode very close to \$54.20, the price specified in the purchase agreement and eventually agreed to. The distribution also indicates meaningful risk of the deal collapsing, leading to large negative returns from the October 21 spot price of \$49.9 per share. The Twitter episode thus provides a clean illustration of how our method recovers a discrete, announcement-specific risk-neutral distribution that concentrates mass on economically meaningful states.

⁶See “Twitter lawsuit halted so Elon Musk can close deal by Oct. 28,” Reuters, October 6, 2022.

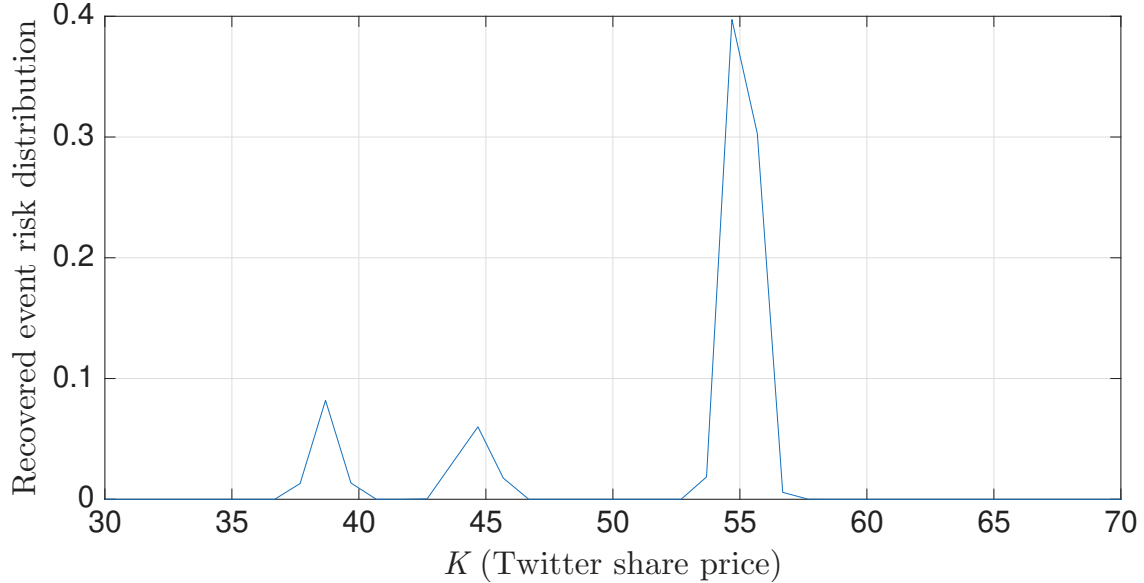


Figure 8: *Recovered event risk distribution for Twitter on October 21, 2022.*

An antitrust court decision: AT&T and Time Warner

Our second example examines the proposed merger between AT&T and Time Warner. AT&T agreed in 2016 to acquire Time Warner for roughly \$80 billion, but the U.S. Department of Justice filed suit to block the deal on antitrust grounds. The case went to trial in a federal district court. The judge in the case stated at the end of April 2018 that he planned to issue a decision on June 12.⁷ The underlying merger agreement was set to expire on June 21, so the court’s guidance effectively created a narrow window in which the key announcement — whether the deal would be allowed to proceed — had to be made.

Figure 9 plots the recovered announcement distribution for Time Warner using options observed on Friday, June 8, expiring on the following two Fridays (June 15 and June 22). As in the Twitter example, this is the last weekly pre-decision trading day before the expected ruling. The distribution again places substantial mass on a price region consistent with the deal being approved (at \$107 per share). It also assigns meaningful probability to negative return outcomes (with prices below the June 8 spot price of \$95.34 per share) that would prevail if the acquisition were blocked or delayed. This pattern is consistent with investors viewing approval as the most likely outcome (as was eventually realized), but not a foregone conclusion. This episode thus gives a second illustration of how our method recovers a discrete, event-specific distribution in a setting where the timing of a legal decision is known *ex ante*.

⁷See “AT&T-Time Warner Trial: Judge Says He’ll Have Decision in June,” *Variety*, April 30, 2018.

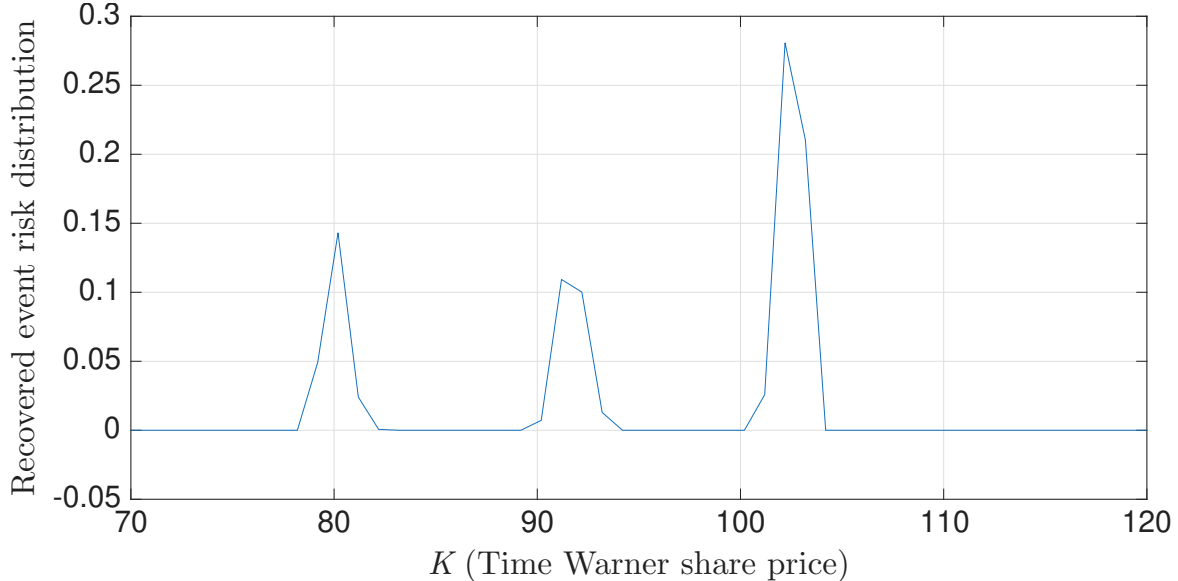


Figure 9: *Recovered event risk distribution for Time Warner on June 8, 2018.*

A systematic event: the Brexit referendum

Our third example is the United Kingdom’s referendum on membership in the European Union, which took place on June 23, 2016. The date of the vote was set months in advance, in February 2016, with a binary outcome (“Remain” in the EU vs. “Leave”). The British pound was widely expected to fall sharply if Leave won. The referendum therefore provides a natural setting in which to implement the two-point recovery from Theorem 3: the pound falls if Leave wins and rises if Remain wins, and the objects to recover are the two moves’ sizes and probabilities. Unlike the two previous examples, the event is systematic, so the risk-neutral distribution we recover need not coincide with the physical one. The question is whether it is nonetheless informative about the market’s assessment of the vote.

We use options on the CurrencyShares British Pound Sterling Trust (ticker FXB), the largest exchange-traded fund tracking the GBP/USD exchange rate, on trading days from May 2 through June 22, 2016 (the last trading day before the vote). The two FXB option expiration dates are July 15 and September 16, 2016. Since both dates are after the vote, this falls under case (ii) of Theorem 3.⁸ On each trading day, we recover the pair of jump sizes and the associated probabilities using the procedure described after Theorem 3. The recovered probability of the down jump is our option-implied probability of a Leave vote.⁹ As a benchmark for comparison, we use the probability of a Leave vote implied by bookmaker odds, which we collect from archived odds comparisons

⁸A benefit in using Theorem 3 over Theorem 1 for recovery in this case is that it places no restriction on the ratio of the two option expiration horizons, t and T . We update option prices day-by-day when implementing the algorithm, using weekly normal-period distributions where the time to expiration is rounded to a whole number of weeks.

⁹To reduce day-to-day noise in the recovered probability, we use two-day moving averages.

on Oddschecker (<https://www.oddschecker.com>) and normalize so that the Leave and Remain probabilities sum to one. These bookmaker odds are available on 21 dates between April 22 and June 22. We also compare the recovered probability to the VIX; given that the Brexit vote was considered an important systematic event, the VIX tracked Brexit odds to some degree in advance of the vote.

Figure 10 plots the two probabilities — the option-implied Brexit probability as recovered from our algorithm, and the bookmaker-implied probability — alongside the VIX. The option-implied and bookmaker probabilities track each other closely: across the trading dates on which both are observed, their correlation is 0.78 in levels and 0.58 in changes. Both average a little above 0.2 in May and rise through the first half of June, peaking on June 14 (at 0.46 for the option-implied probability and 0.39 for the bookmaker probability) before falling over the final week. The VIX, by contrast, stays flat near 15 until June 10 and then jumps to 21 on June 13; across the common trading dates, its correlation with the bookmaker probability is 0.60 in levels and 0.17 in changes. The option-implied probability of a Leave vote begins to rise on June 6, several days before the VIX does, and it tracks the betting odds far more closely.¹⁰

Averaged across the trading days in our sample, the recovered FXB jump size is +3.8% if Remain wins and −10.9% if Leave wins (not plotted in the figure), with fairly little variation in these values over time. Leave won on June 23, and FXB fell 8.4% the next day, close to both the average implied decline and the implied decline as of June 22 (−9.1%). The method accordingly appears to work well for tracking both implied probabilities and jump size in this example.

3.2 Earnings announcement data

To broaden the analysis, we now study quarterly earnings announcements for large U.S. firms in the S&P 500. We test whether our recovered ex ante announcement distribution helps predict the distribution of returns on the day of the earnings announcement. Our sample covers earnings announcements over the 2010s.¹¹ We obtain quarterly earnings announcement dates from Compustat and match these to daily announcement-day stock returns from CRSP. Option quotes come from OptionMetrics Ivy DB US.

We set one model period to be one week, to avoid confounding from other discrete risks that might arise with longer horizons. Weekly options expire on Fridays at market close. For each earnings announcement, we therefore identify three Fridays. The first is the Friday before the announcement, which we treat as trading date $t = 0$. The second ($t = 1$) is the first Friday after

¹⁰The general recovery method of Theorem 2, applied to the same FXB options following the steps described in the subsections below (treating the two expirations as one and two periods ahead), tracks the bookmaker probability closely as well: the recovered probability of an FXB return below −5% has a correlation of 0.89 in levels and 0.88 in changes with the bookmaker probability across the same dates.

¹¹Firm-level weekly options were not generally traded in 2010–2012. Our analysis considers a weekly horizon, so our sample effectively starts in 2013 and runs through the end of 2019.

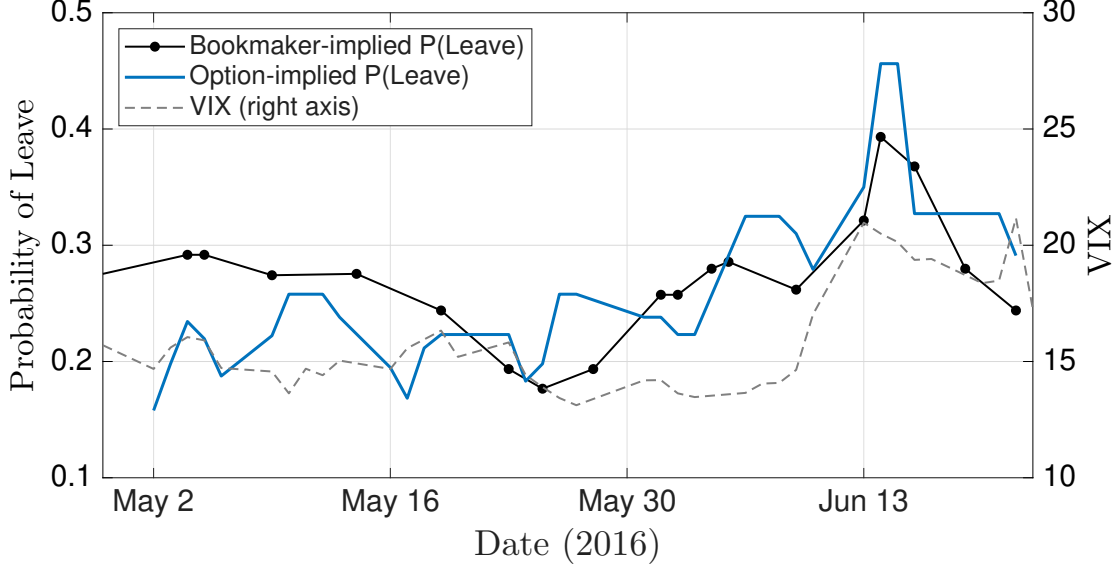


Figure 10: *Option-implied and bookmaker-implied probabilities of a Leave vote ahead of the June 23, 2016 Brexit referendum, with the VIX on the right axis.*

the announcement,¹² and the third ($t = T = 2$) is the following Friday. On the pre-announcement trading day, we collect all options that expire on either of the two post-announcement Fridays. This gives us a pre-announcement option panel with two nearby maturities, for which the announcement occurs between the trading date and $t = 1$.

We apply a set of simple filters to the option data: we require a strictly positive bid price, a strictly positive Black-Scholes implied volatility, and strictly positive open interest or trading volume. We work with mid price quotes. For each expiry we require at least eight distinct strikes with valid quotes. Events that do not meet these criteria are excluded. Applying these filters yields nearly 4,300 earnings events with usable pre-announcement options over our sample period. This is a subset of the universe of nearly 20,000 S&P 500 earnings announcements over the 2010s, as many firms do not have sufficient (if any) option quotes at the one- or two-week horizon. This particularly affects small stocks and early-sample observations.

3.3 Smoothing and convexification of noisy price observations

The results in Section 2, in particular equations (13)–(17), show how to recover the announcement distribution from option prices. To implement this empirically, we first face the issue that observed single-stock option prices have meaningful measurement noise. Because the recovery algorithm ultimately takes second differences of put prices, small pricing errors can generate large oscillations in the implied distribution. These issues are alleviated but not completely addressed by the approach in Theorem 2. We therefore smooth the raw option quotes and impose the convexity restriction

¹²A small subset of announcements happen on Friday. In all cases, these occur before the market open, so for these we use the announcement date as the first expiration date.

that holds for no-arbitrage put-price schedules before applying Theorem 2.

Our smoothing procedure operates as follows. First, for each maturity, we start from the Black-Scholes implied volatilities for puts and calls reported by OptionMetrics. As out-of-the-money options are typically more liquid than in-the-money options, we use puts at strike prices K below the current underlying price S_0 , and calls at strikes $K > S_0$. To avoid a discrete jump in the implied volatility and price schedule at $K = S_0$, we blend the put and call observations in a symmetric band $[K_\ell, K_h]$ around S_0 . In our implementation we set $K_\ell = 0.9S_0$ and $K_h = 1.1S_0$, and within this range, we consider all available strikes for which both put and call implied volatilities are available.

Formally, denote the put and call implied volatilities as $\sigma^P(K)$ and $\sigma^C(K)$, and denote our final blended implied volatility as $\sigma^{\text{blend}}(K)$. For strikes $K < K_\ell$, we take $\sigma^{\text{blend}}(K) = \sigma^P(K)$ whenever put implied volatilities are available.¹³ For strikes $K > K_h$, we take $\sigma^{\text{blend}}(K) = \sigma^C(K)$ whenever call implied volatilities are available. On strikes in the band $[K_\ell, K_h]$ where both puts and calls have valid implied volatilities, we combine the two according to

$$\sigma^{\text{blend}}(K) = w(K) \sigma^P(K) + (1 - w(K)) \sigma^C(K), \quad (20)$$

where $w(K) = 1$ for $K \leq K_\ell$, $w(K) = 0$ for $K \geq K_h$, and $w(K) = \frac{K_h - K}{K_h - K_\ell}$ for $K_\ell < K < K_h$. Puts thus dominate in the left tail and calls dominate in the right tail, with a smooth linear transition near at-the-money strikes where both sides trade actively. Using $\sigma^{\text{blend}}(K)$, the spot price S_0 , the time to maturity, and the risk-free yield curve, we then evaluate the Black-Scholes put-price formula to obtain the no-arbitrage price $P(K) = P_K$ at each strike. The collection $\{K_j, P(K_j)\}_{j=1}^J$ serves as the input to the next step.

Second, to ensure positive risk neutral probabilities, we convexify the put-price schedule with respect to strike price. Fix an expiry and write $\{K_j, P_j\}_{j=1}^J$ for the strikes and observed (blended) put prices after filtering. We fit a least-squares quadratic spline approximation $\tilde{P}(K)$ to the observed put prices,

$$\min_{\tilde{P}} \sum_{j=1}^J \left(\tilde{P}(K_j) - P_j \right)^2, \quad (21)$$

subject to the convexity restriction

$$\tilde{P}''(K) \geq 0 \quad \text{for all } K. \quad (22)$$

In addition, we impose that the risk-neutral probabilities implied by the double difference of $\tilde{P}$ sum to one by enforcing that the terminal value of the cumulative distribution function associated with $D_+ D_- \tilde{P}$ equals one on a fine strike grid, in line with the construction in equations (15)–(17). This

¹³In implementing Theorem 2, we consider strike prices with moneyness ranging from -40% to $+40\%$. When this range is wider than the range of strikes with observable quotes, we impute implied volatility from the lowest available put strike for the left tail and from the highest available call strike for the right tail.

constrained spline approximation problem is a quadratic program in the spline coefficients. This projection replaces the noisy quotes by the closest smooth, convex curve with respect to strike price, such that the associated risk-neutral distribution is positive and integrates to one. It eliminates no-arbitrage violations and reduces the microstructure noise that would otherwise be magnified when taking second differences of prices in the last step of the procedure. We apply this convexification separately to each maturity.

Finally, we use the convexified put prices for the two post-announcement expiration dates to implement the algorithm in Theorem 2. Let $\tilde{P}^1$ and $\tilde{P}^2$ denote the fitted put-price functions for the shorter and longer maturity, respectively. We evaluate these functions on an evenly spaced grid of strikes to construct the discrete vectors P^1 and P^2 in (15)–(16), which we then use to obtain the announcement distribution $p = (p_k)_k$ through the mapping in (17). In the final step we again apply our convexified-spline approximation to the put-price schedule associated with p , which ensures that the implied state prices remain nonnegative and that small oscillations in the recovered announcement distribution are eliminated. Our final announcement distribution is over a range of strikes with moneyness between -40% and $+40\%$, spaced \$1 apart from each other.

For each event, we then compute the variance, skewness, kurtosis, and crash probability implied by the ex ante recovered announcement distribution. The crash probability is defined as the probability that the announcement-day return will fall below the 5th percentile of all earnings-announcement realized returns; in our sample, this corresponds to an announcement-day return of -6.0% (non-annualized).

3.4 Summary statistics and earnings announcement return predictability

Table 1 reports summary statistics for the announcement-day equity returns and for features of the recovered announcement distributions. Our baseline sample consists of 4,288 earnings announcements. Realized announcement returns tend to be volatile, with a daily standard deviation of above 4% for non-annualized returns. This corresponds to realized variance of 0.002. In comparison, the ex ante variance implied by the recovered announcement distribution is slightly higher, at 0.003.¹⁴ The model-implied probability of a crash below -6% (the empirical 5th percentile for announcement returns) is 8.3%, slightly elevated relative to the true crash frequency of 5%.

We next relate the ex ante objects to realized announcement outcomes. We start by regressing the centered squared announcement-day return on the recovered announcement distribution variance. We benchmark the predictive power of our recovered measure by comparing it to a prediction based on the one-week implied variance for that stock, constructed identically to the one-week squared VIX but at the single-stock level. We refer to this implied variance as VX^2 ,

¹⁴In the analysis below, we also use higher moments of the announcement distribution. On average, these recovered distributions feature modest negative skewness and positive excess kurtosis, as in the data for realized returns.

	Mean	Median	Std. dev.	Obs.
Realized announcement return (%)	0.12	0.27	4.24	4,288
Ex ante announcement variance	0.0033	0.0021	0.0035	4,288
Implied crash probability (%)	8.34	4.89	11.07	4,288

Table 1: *Summary statistics for earnings announcements.*

Notes: Returns are daily earnings-announcement-day equity returns (non-annualized). Announcement variance is the ex-ante variance of the recovered announcement distribution, measured in squared-return units. Implied crash probability is the model-implied probability that the announcement return falls below -6% (the empirical 5th percentile).

following the CBOE’s convention for referring to single-stock volatility indices.¹⁵ All regressions cluster standard errors by ticker and calendar quarter. Results are shown in Table 2.

Column (1) of Table 2 shows that the recovered announcement-distribution variance is a strong and significant predictor of centered squared returns. Each unit increase in ex ante variance corresponds to a 0.46-unit increase in the centered squared announcement-day return on average. This estimate likely reflects some degree of attenuation bias, given the noise in the option prices and in the announcement distribution recovered from them, but overall the results indicate strong predictability. This predictability is comparable to the predictability in centered squared returns obtained from using the squared VX at the one-week horizon, as in column (2). In a joint regression (column (3)), the announcement-distribution variance remains a significant predictor, while the coefficient on VX^2 falls to 0.08 and is no longer distinguishable from zero.

Table 3 considers predictions for other aspects of the realized return distribution. Column (1) regresses the centered third moment of the announcement-day return on the ex ante skewness implied by the recovered announcement distribution; column (2) regresses the centered fourth moment of the realized return on the ex ante kurtosis; and column (3) regresses a crash indicator (equal to 1 for a realized return below the 5th percentile of the return distribution, -6%) on the recovered ex ante crash probability. The announcement-implied skewness has little predictive power for the third moment, but announcement-implied kurtosis loads positively on the realized fourth moment, indicating that fatter ex-ante tails are associated with fatter realized tails. The ex-ante crash probability is also positively related to realized crashes, showing that our recovered distribution helps predict crash risk.

The results indicate that our method provides useful ex ante information about announcement-specific risk. It also removes the normal-period confounding that affects measures like implied volatility or the VX, as reflected in our measure’s strong performance relative to the VX in Table 2.

¹⁵We construct VX as of the same trading date at which we calculate the announcement distribution, using the observed option prices expiring the Friday following the announcement. Our VX calculation follows the CBOE’s VIX methodology, described at <https://www.cboe.com/tradable-products/vix/faqs/>. For an example of a single-stock VX index, see <https://www.cboe.com/us/indices/dashboard/vxap1/>.

	(1)	(2)	(3)
Announcement variance	0.4624*** (0.0932)		0.3845*** (0.1375)
VX ²		0.3192*** (0.0651)	0.0826 (0.0946)
Obs.	4288	4288	4288
R ²	0.065	0.047	0.066

Table 2: *Regressions for centered squared realized announcement-day returns.*

Notes: Dependent variable is the announcement-day centered squared return $(r - \mu)^2$, where μ is the mean of the ex ante announcement distribution. “Announcement variance” is the ex ante announcement distribution variance. “VX²” is the squared pre-announcement one-week single-stock volatility index, constructed using the VIX formula. The sample uses all earnings announcements with available options. Standard errors are two-way clustered by ticker and calendar quarter. Statistical significance at the 10%, 5%, and 1% levels denoted by *, **, and ***, respectively.

	(1) Third moment	(2) Fourth moment	(3) Crash indicator
Corresponding ex ante recovered moment	-0.0514 (0.1064)	0.1925** (0.0855)	0.2015*** (0.0476)
Obs.	4288	4288	4288
R ²	0.000	0.016	0.009

Table 3: *Regressions for higher moments and crash indicators.*

Notes: Dependent variables are the centered third and fourth moments of the announcement-day return and an indicator for an announcement-day crash (return below -6% , the 5th percentile of the realized announcement return distribution). Each column regresses the corresponding realized outcome on the matching announcement-implied moment: skewness, kurtosis, and the ex-ante crash probability. We use all events with available options. Standard errors are two-way clustered by ticker and calendar quarter. Statistical significance at the 10%, 5%, and 1% levels is denoted by *, **, and ***, respectively.

3.5 Predicting post-announcement distributions

We empirically test how well our method predicts post-announcement distributions (see the third example in Section 2.5). For each earnings announcement in our sample, we compare the recovered q to the ex-post risk-neutral distribution q^{post} , obtained from options traded on the first post-announcement Friday for the second Friday — that is, the option-implied density for a week that does not span any announcement. As benchmarks, we use the two naive alternatives described above: the one-week density q^1 (which conflates announcement and normal-period risk) and the “square root” of q^2 (which assumes identical distributions each week).¹⁶

¹⁶All distributions — forecasts and ex-post truth — are centered to zero mean on the comparison grid before scoring. This removes small location differences arising from the use of different forward prices for the pre-announcement and

One practical issue is that our recovery method involves pointwise division in the Fourier domain, which amplifies noise at frequencies where the denominator is near zero. To address this, we apply Tikhonov regularization — formally identical to ridge regression — which damps high-frequency noise while preserving the low-frequency structure of the recovered density.¹⁷ Without regularization, many Fourier coefficients of $\hat{q}^1(\omega)$ are near zero, producing recovered densities with substantial negative mass that are then distorted by the convexification step.

Table 4 reports results for $N = 3,735$ events.¹⁸ The key comparison is the variance regression: $\sigma_{\text{post}}^2 = \alpha + \beta \hat{\sigma}^2 + \varepsilon$, where σ_{post}^2 is the variance of q^{post} and $\hat{\sigma}^2$ is the variance of the forecast density. A perfect forecast implies $\beta = 1$ and $R^2 = 1$. The regularized recovered distribution explains substantially more cross-sectional variation in post-announcement variance than either benchmark: $R^2 = 0.19\text{--}0.22$, compared to 0.09 for the one-week density and 0.04 for the convolution square root. The regression slope is also closer to unity ($\beta = 0.48\text{--}0.54$ versus 0.24–0.25), indicating a better-calibrated forecast of event-specific variance. The Kolmogorov–Smirnov (KS) statistic tells a similar story: the regularized recovery achieves a mean KS of 0.139–0.143, compared to 0.156 for q^1 and 0.170 for the “square root” of q^2 . The unregularized inversion performs worst on both the quadratic score and KS statistic, confirming that regularization is essential to yield useful distributional forecasts.

These results are consistent with the interpretation that our recovery method successfully extracts normal-period risk-neutral information that is distinct from the raw option-implied densities contaminated by announcement risk. Note that the same Tikhonov regularization can also be applied to the recovered announcement distribution p , where a qualitatively similar noise-amplification issue can arise from near-zero Fourier coefficients.¹⁹

post-announcement densities, and ensures that the comparison focuses on distributional shape and spread rather than location. We evaluate forecasts on a common moneyness grid $m \in [-0.20, 0.20]$ with spacing $\Delta m = 0.005$ (81 points).

¹⁷Specifically, we replace the pointwise ratio $\hat{q}^2(\omega)/\hat{q}^1(\omega)$ with the regularized ratio $\hat{q}^2(\omega) \overline{\hat{q}^1(\omega)} / (|\hat{q}^1(\omega)|^2 + \gamma)$, where $\gamma = \gamma_0 \cdot |\hat{q}^1(\omega)|^2$ and γ_0 is a scalar penalty parameter. The analogy to ridge regression is exact: γ_0 plays the role of the penalty on regression coefficients, and $|\hat{q}^1(\omega)|^2 + \gamma$ replaces the near-singular “moment matrix” $|\hat{q}^1(\omega)|^2$. We select γ_0 by evaluating a grid of penalty values on a subsample of the scoring events (every tenth event), comparing the recovered normal-period densities to the ex-post distributions. The best-performing values are $\gamma_0 \approx 0.5\text{--}1.0$, depending on the scoring criterion, and Table 4 reports results for both. We also considered Lavrentiev regularization, which adds a real constant c directly to the denominator — replacing $\hat{q}^1(\omega)$ with $\hat{q}^1(\omega) + c$ — rather than regularizing via the squared modulus. This approach performed consistently worse on the criteria reported below, likely because adding a real constant to complex-valued Fourier coefficients shifts them rather than uniformly damping their influence.

¹⁸This sample is smaller than the full earnings announcement sample because the validation requires sufficient post-announcement option data. Events without adequate post-announcement option data are excluded.

¹⁹Applying Tikhonov regularization to the announcement distribution raises the R^2 for variance prediction modestly, and also raises the R^2 for kurtosis more substantially, to 0.152.

Method	Quad. Score	KS	Var. MSE ($\times 10^6$)	β	R^2
q (Tikhonov, $\gamma_0=0.5$)	-0.000271	0.139	3.35	0.537	0.191
q (Tikhonov, $\gamma_0=1.0$)	-0.000222	0.143	4.16	0.483	0.224
q (unregularized)	-0.000776	0.221	3.92	0.547	0.112
One-week density (incl. annnc.)	-0.000176	0.156	8.48	0.250	0.086
$\sqrt{q^2}$ (i.i.d. assumption)	-0.000277	0.170	9.66	0.235	0.042

Table 4: *Normal-period distribution: comparison with ex-post risk-neutral distribution.*

Notes: $N = 3,735$ scoring events. All distributions are centered to zero mean before scoring. The regularization parameter γ_0 is selected by evaluating a grid of penalty values on a subsample of the scoring events; results are reported for the two best-performing values. Quadratic score: higher is better. KS: Kolmogorov–Smirnov statistic (lower is better). Variance regression: $\sigma_{\text{post}}^2 = \alpha + \beta \hat{\sigma}^2 + \varepsilon$, where σ_{post}^2 is the variance of the ex-post risk-neutral distribution and $\hat{\sigma}^2$ is the variance of the forecast density.

4 Conclusions

We have introduced a nonparametric method for recovering event risk distributions from option prices at two maturities. In simulations, the method recovers the announcement distribution with essentially zero errors. Empirically, the recovered announcement distribution is a strong predictor of realized announcement-day return moments, outperforming conventional option-implied measures that conflate announcement and normal-period risk. The decomposition also yields a normal-period risk-neutral distribution that substantially outperforms naive benchmarks — including the one-week option-implied density and the convolution square root of the two-week density — at predicting the post-announcement risk-neutral distribution. Tikhonov regularization further improves recovery quality for both the normal-period and announcement distributions.

Beyond earnings announcements, our method applies naturally to any scheduled event with options at two expirations, at least one of them after the event. FOMC announcements, and their effect on the market, have received substantial attention in recent years. FOMC events are hardly idiosyncratic, so an additional challenge here is that our method recovers the risk-neutral announcement distribution, whereas one is typically interested in the objective announcement distribution. Our method may still be informative, especially with respect to higher moments of these distributions.

Another future application is to prediction markets, where rapidly expanding companies like Polymarket, Kalshi, and DraftKings now structure their products as binary options that can be traded in a market. It is an open question how informationally efficient these new markets are — concerns over insider trading have been voiced, for example. Our method could be used to recover implied event distributions from option prices for events that are known to affect the market, such as elections, and to compare those distributions with the probabilities implied by prices in the corresponding prediction markets, to study whether one is more informative than the other.

Appendices

A The general model

A.1 Sequence spaces

Let

$$\mathbb{R}^{\mathbb{Z}} = \{x : \mathbb{Z} \rightarrow \mathbb{R}\}.$$

We use the Banach spaces

$$\begin{aligned}\ell^p &= \left\{ x \in \mathbb{R}^{\mathbb{Z}} : \sum_{k \in \mathbb{Z}} |x_k|^p < \infty \right\}, \quad p \geq 1, \\ \ell^\infty &= \left\{ x \in \mathbb{R}^{\mathbb{Z}} : \sup_{k \in \mathbb{Z}} |x_k| < \infty \right\},\end{aligned}$$

with norms

$$\|x\|_p = \left(\sum_k |x_k|^p \right)^{1/p}, \quad \|x\|_\infty = \sup_k |x_k|.$$

We also define ℓ^0 as the subspace of ℓ^1 with finite support, and ℓ_K^0 as the subspace whose elements satisfy $x_k = 0$ for $|k| > K$.

For $1 \leq p < q$, we have $\ell^p \subset \ell^q$, and therefore

$$\ell_0^0 \subset \ell_1^0 \subset \ell_2^0 \subset \dots \subset \ell^0 \subset \ell^p \subset \ell^q \subset \ell^\infty \subset \mathbb{R}^{\mathbb{Z}}.$$

Specific elements are $\delta_k \in \ell_k^0$, with $(\delta_k)_k = 1$, and $(\delta_k)_{k'} = 0$, $k \neq k'$. We identify the transition probabilities p and q with elements in ℓ_K^0 , $p = (\dots, p_K, p_{K-1}, \dots, p_{-K}, \dots) \in \ell_K^0$, $q = (\dots, q_K, q_{K-1}, \dots, q_{-K}, \dots) \in \ell_K^0$. The special case when there is no announcement risk arises when $p = \delta_0$. The standard binomial tree model is a special case of our model, in which no announcements occur, $K = 1$, and only the normal-period risk neutral probabilities q_{-1} and $q_1 = 1 - q_{-1}$ are nonzero.

A.2 State space, preferences and pricing

Time is discrete, except for one non-integer time $u > 0$ at which an announcement occurs that changes market beliefs about the asset's payoff. Thus

$$t \in \{0, 1, 2, \dots, t^*, u, t^* + 1, t^* + 2, \dots\}, \quad t^* < u < t^* + 1.$$

The interval $(t^*, t^* + 1)$ is the *announcement period*. All other intervals $(t, t + 1)$, $t \neq t^*$, are *normal periods*. We assume the event time u is known.

The state at date t is

$$s_t = (a_t, m_t) \in \mathbb{Z} \times \mathbb{Z},$$

where a_t summarizes firm-specific, payoff-relevant information for the asset and m_t summarizes aggregate information relevant for the pricing kernel.

A representative agent has expected utility preferences

$$EU_t = E_t \left[\sum_{v=t}^T \rho^{v-t} U(\tilde{C}_v) \right],$$

where $\rho \in (0, 1)$ is the subjective discount factor, U is increasing and strictly concave, and aggregate consumption is

$$\tilde{C}_t = C_t(m_t).$$

The announcement time u is a pure information arrival and is therefore not included in the summation.

Consider a contingent claim with terminal payoff

$$\tilde{F}_T = F(a_T),$$

where the payoff depends only on the payoff-relevant state at time T . Standard equilibrium arguments (see Lucas, 1978) imply that its time- t price is

$$P_t = E_t \left[\rho^{T-t} \frac{U'(\tilde{C}_T)}{U'(\tilde{C}_t)} \tilde{F}_T \right]. \quad (23)$$

Although the terminal payoff depends only on a_T , prices before T may also depend on m_t through the pricing kernel. An equivalent risk-neutral representation is

$$P_t = E_t^{\mathbb{Q}} \left[\left(\prod_{r=t}^{T-1} R_{r,r+1}^{-1} \right) \tilde{F}_T \right], \quad (24)$$

where $R_{r,r+1}$ is the one-period gross risk-free rate and $\mathbb{Q}$ is the risk-neutral measure. We assume the risk-free rate is non-stochastic, so

$$P_t = \left(\prod_{r=t}^{T-1} R_{r,r+1}^{-1} \right) E_t^{\mathbb{Q}} [\tilde{F}_T]. \quad (25)$$

A.3 Normal periods versus announcement periods

The model has two types of state transitions.

Normal periods. During a normal period $(t, t+1)$, the state evolves as

$$(a, m) \mapsto (a + k, m + k), \quad |k| \leq K^N,$$

where $K^N > 0$ is a strictly positive constant. Thus payoff-relevant and pricing-kernel-relevant information move together in normal periods, and moreover may not move more than K^N steps in

either direction. The risk-neutral probabilities for these moves are

$$\mathbb{Q}(s_{t+1} = (a + k, m + k) \mid s_t = (a, m)) \equiv q_k. \quad (26)$$

We assume these transition probabilities are time-invariant and state-independent,²⁰ so the risk-neutral state transition behaves like a time-invariant Markov process.

Announcement period. During the announcement period $(t^*, t^* + 1)$, the state changes in two steps. First, at the announcement time u , the state evolves as

$$(a, m) \mapsto (a + k, m), \quad |k| \leq K^p,$$

with

$$\mathbb{Q}(s_u = (a + k, m) \mid s_{t^*} = (a, m)) = \mathbb{P}(s_u = (a + k, m) \mid s_{t^*} = (a, m)) \equiv p_k, \quad (27)$$

where $\sum_k p_k = 1$ and $K^p \geq 0$. The case $K^p = 0$ corresponds to a no-news announcement, so that $p_0 = 1$.

The announcement moves only the payoff-relevant state. It does not affect the pricing-kernel state m , so physical and risk-neutral announcement probabilities coincide. Economically, the event is firm-specific and has negligible implications for aggregate consumption.

Second, at (or right before) $t^* + 1$, the economy again evolves according to the normal-period transition probabilities q_k . Hence the distribution $\{q_k\}$ captures ordinary background news, while $\{p_k\}$ captures the discrete announcement risk that we seek to recover.

A.4 Laurent operators

Both normal and announcement period events can be modeled by Laurent operators $\mathcal{N} : \mathbb{R}^{\mathbb{Z}} \rightarrow \mathbb{R}^{\mathbb{Z}}$ and $\mathcal{A} : \mathbb{R}^{\mathbb{Z}} \rightarrow \mathbb{R}^{\mathbb{Z}}$, acting on the payoff-relevant part of the state (the projection of $(a, m) \in \mathbb{Z} \times \mathbb{Z}$ onto $a \in \mathbb{Z}$), defined by

$$(\mathcal{N}x)_n = \sum_k x_k q_{n-k}, \quad (\mathcal{A}x)_n = \sum_k x_k p_{n-k}.$$

Here, $(\mathcal{N}x)_n$ is the probability of being in payoff-relevant state $a + n$ at $t + 1$, given that the probability distribution of a is x at t ; and similarly for the announcement operator $\mathcal{A}$. Technically, the operator $\mathcal{N}$ represents the forward Chapman-Kolmogorov operators for a normal transition, and $\mathcal{A}$ for an announcement transition. The sequence of events and corresponding Laurent operators are shown in Figure 11. This is the infinite-dimensional equivalent of the finite dimensional version shown in Figure 3 in the body of the paper.

Since Laurent operations are convolution operators, the two operators commute:

$$\mathcal{N}\mathcal{A} = \mathcal{A}\mathcal{N}.$$

²⁰This may seem like a restrictive specification, but our applications will be over fairly short time periods, say $t \leq 4$, where t is measured in hours, days, weeks, or months, in which case such a time invariance assumption may be reasonable.

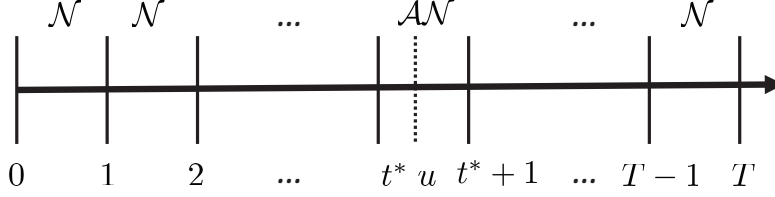


Figure 11: Sequence of information events and corresponding Laurent operators.

Their adjoint operators are

$$(\mathcal{N}^*x)_n = \sum_k x_k q_{k-n}, \quad (\mathcal{A}^*x)_n = \sum_k x_k p_{k-n},$$

representing the backward Chapman-Kolmogorov operators. These operators also commute:

$$\mathcal{N}^* \mathcal{A}^* = \mathcal{A}^* \mathcal{N}^*.$$

Technically, the adjointness relation is defined on the subspace ℓ^2 , with inner product

$$\langle v, w \rangle = \sum_k v_k w_k.$$

Since both physical and risk-neutral probabilities sum to one, the constant vector

$$\mathbf{1} = (\dots, 1, 1, 1, \dots) \in \ell^\infty$$

is an eigenfunction with the eigenvalue one for both the operators $\mathcal{N}^*$ and $\mathcal{A}^*$, and since these operators are convolution operators the relations also hold for $\mathcal{N}$ and $\mathcal{A}$:

$$\mathbf{1} = \mathcal{N}\mathbf{1}, \quad \mathbf{1} = \mathcal{A}\mathbf{1}. \tag{28}$$

A.5 Recovering the announcement distribution from state prices

Consider the initial state $s_0 = (0, 0)$, and let δ_k^t denote the Arrow-Debreu security that pays one dollar in payoff-relevant state k at time t , for $t = 1, \dots, T$. Let its date-0 price be $[\Psi^t]_k$, so that $\Psi^t \in \mathbb{R}^Z$. Define the associated risk-neutral probabilities by

$$q^t = \left(\prod_{r=0}^{t-1} R_{r,r+1} \right) \Psi^t.$$

Thus q^t is the date-0 risk-neutral distribution over payoff states at horizon t .

If the announcement occurs after date t , then q^t reflects only normal transitions. If the announcement occurs before date t , then q^t reflects both normal transitions and the announcement shock. This is exactly what creates the identification problem. With multiple maturities, it is also what makes identification possible.

Let $f \in \mathbb{R}^{\mathbb{Z}}$ be the payoff vector with components $f_k = F(k)$. Define

$$v_t = \frac{1}{R^{T-t}} (A^*)^{\iota(t)} (N^*)^{T-t} f, \quad \iota(t) = \begin{cases} 1, & t < u, \\ 0, & t > u. \end{cases}$$

Then the pricing relation can be written as

$$P_t = [v_t]_{a_t}.$$

It follows that

$$v_t = \frac{1}{R_{t,t+1}} \mathcal{N}^* v_{t+1}, \quad t \neq t^*, \quad \text{and} \quad v_{t^*} = \frac{1}{R_{t^*,t^*+1}} \mathcal{A}^* \mathcal{N}^* v_{t^*+1}. \quad (29)$$

In the special case in which the payoff-relevant state itself indexes the time-invariant price of an asset, so that

$$P_t = G(a_t), \quad t = 0, 1, \dots, T,$$

for some $G \in \mathbb{R}^{\mathbb{Z}}$, we have $v_t = v_{t+1} \equiv v$. If in addition the risk-free rate is constant, $R_{t,t+1} \equiv R$, then (29) implies

$$v = \frac{1}{R} \mathcal{N}^* v, \quad v = \mathcal{A}^* v. \quad (30)$$

This is the case for the underlying asset in the classical binomial tree.

A.6 Date-0 state-price distributions

Let Ψ^t denote the date-0 Arrow–Debreu (AD) price vector at horizon t , and let

$$q^t = \left(\prod_{r=0}^{t-1} R_{r,r+1} \right) \Psi^t$$

be the corresponding risk-neutral distribution for the t -horizon AD securities (at date 0). Then

$$q^t = (\mathcal{A})^{1-\iota(t)} (\mathcal{N})^t \delta_0, \quad \iota(t) = \begin{cases} 1, & t < u, \\ 0, & t > u. \end{cases} \quad (31)$$

If $u > 1$, then $q_k^1 = q_k$ for all k . Also, if $q \in \ell_{K^N}^0$ and $p \in \ell_{K^p}^0$, then $q^t \in \ell_{tK^N + K^p}^0$.

A.7 Recovering the announcement probability distribution, p

To recover the announcement risk distribution, we use the theories of Laurent polynomials, Z -transforms, and Fourier transforms. Specifically, the systems of equations stated in the main body of the paper (5,6) are in general large and highly nonlinear. But the fact that $\mathcal{A}$ and $\mathcal{N}$ are convolution operators of finite length allows us to analyze and solve these systems efficiently.

Specifically, for a general probability vector $f \in \ell_K^0$, define the Laurent polynomial (which is a

rational function)

$$F(z) = \mathcal{Z}[f](z) \stackrel{\text{def}}{=} \sum_{k=-K}^K f_k z^{-k}.$$

Here, $\mathcal{Z}[f]$ defines the Z -transform of f . By the Cauchy residue theorem from complex analysis, it follows that the coefficients f_k can be recovered by the integration formula

$$f_k = \frac{1}{2\pi i} \oint_C F(z) z^{k-1} dz.$$

Here, C is a counterclockwise contour in the complex plane that encloses the origin and all poles of $F(z)$.

The Laurent polynomials associated with the normal and announcement event operators are then:

$$A(z) = \sum_k p_k z^{-k}, \quad \text{and} \quad Q(z) = \sum_k q_k z^{-k}.$$

It follows, again from the theory of Z transforms, that if we define

$$\begin{aligned} X(z) &= \mathcal{Z}[q^r](z) = \sum_k q_k^r z^{-k}, \\ V(z) &= \mathcal{Z}[q^{T-r}](z) = \sum_k q_k^{T-r} z^{-k}, \\ W(z) &= \mathcal{Z}[q^T](z) = \sum_k q_k^T z^{-k}, \end{aligned}$$

then

$$\begin{aligned} X(z) &= A(z)^{\iota(r)} Q(z)^r, \\ V(z) &= A(z)^{\iota(T-r)} Q(z)^{T-r}, \\ W(z) &= A(z) Q(z)^T, \end{aligned}$$

This allows us to prove Theorem 1.

Proof of Theorem 1: From the above formulas, it follows that when $u > r$,

$$\frac{W(z)}{X(z)^x} = \frac{A(z) Q(z)^T}{(Q(z)^r)^x} = A(z),$$

and when $u < T - r$,

$$\frac{V(z)^x}{W(z)^{x-1}} = \frac{(A(z) Q(z)^{T-r})^x}{(A(z) Q(z)^T)^{x-1}} = \frac{A(z)^x Q(z)^{(T-r)x}}{A(z)^{x-1} Q(z)^{T(x-1)}} = A(z).$$

By the Cauchy residue theorem, and the fact that $A(z)$'s only pole (of multiplicity at most K_p) lies at the origin, $z = 0$, it follows that $p_k = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{ik\omega} A(e^{ik\omega}) d\omega$, completing the proof.

Alternatively, the theory of Fourier transforms can be used to understand the result. It is a

standard result that the Fourier transform,

$$\hat{p}(\omega) = \mathcal{F}[p](z) = \sum_{k=-K}^K p_k e^{-i\omega},$$

maps ℓ_K^0 into the space $L^2(-\pi, \pi)$, with the corresponding unique inverse transform

$$p_k = \mathcal{F}^{-1}[\hat{p}](k) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{ik\omega} \hat{p}(\omega) d\omega.$$

■

The impossibility in general to recover p from only one date's option prices is easy to see by studying the transformed space. Consider, e.g., the case when $u < t$, so that the announcement occurs before the first expiration date. In the transformed space, $X(z) = A(z)Q(z)^r$ is observed. In general, the product of two functions is not sufficient to recover its two constituents. This is the case here too, even if one takes into account that both $A(z)$ and $Q(z)^r$ are Laurent polynomials. Specifically, by the fundamental theorem of algebra a unique factorization of $X(z)$ exists

$$X(z) = \beta z^{-K_p - rK_q} (z - \alpha_1)^{\gamma_1} (z - \alpha_2)^{\gamma_2} \cdots (z - \alpha_v)^{\gamma_v},$$

for some v , and coefficients $\beta \in \mathbb{R}$, $\alpha_1, \dots, \alpha_v \in \mathbb{C}$, and $\gamma_1, \dots, \gamma_v \in \mathbb{N}$. Additional information is then needed to know which of these factors should be attributed to $A(z)$. This is what happened in the simple setting in Section 2.1, in which we identified multiple solutions.

B Using the Fast Fourier Transform, FFT

For $f \in \ell_K^0$, define the FFT by

$$\hat{f}_\omega = \mathcal{F}[f](\omega) = \sum_{k=-K}^K f_k e^{-2\pi i k \omega / (2K+1)}, \quad \omega = -K, -K+1, \dots, K. \quad (32)$$

The inverse FFT is then defined by

$$f_k = \mathcal{F}^{-1}[\hat{f}](k) = \frac{1}{2K+1} \sum_{\omega=-K}^K \hat{f}_\omega e^{2\pi i k \omega / (2K+1)}, \quad k = -K, -K+1, \dots, K. \quad (33)$$

Note that

$$\mathcal{F}[f](\omega) = \mathcal{Z}[f]\left(e^{-2\pi i \omega / (2K+1)}\right),$$

which makes the FFT useful for numerically recovering p . Indeed, FFT may be viewed as a numerical quadrature (i.e., integration) formula for the inverse Fourier transform, which given the aforementioned link between the Fourier transform and Z -transform discussed in the proof of Theorem 1, is useful.

C Proofs

Proof of Theorem 1: See Appendix A.

Proof of Theorem 2: First, note that since $P^2 \in \ell_K^0$, and $q^2 \in \ell_{2K^N+K^p}^0$, it follows that $2K^N + K^p \leq K$. This in turn implies that $q_{-K} = 0$, and $p_{-K} = p_{-K+1} = 0$.

Now, the translation properties of the FFT imply that $\mathcal{F} \left[\widehat{D_+ D_-} f \right] = 2(\cos(\omega) - 1) \hat{f}(\omega)$, $\omega \neq 0$, for any $f \in \ell_K^0$.

It is straightforward to verify that $\dot{P}^i$ is constructed so that $\left(\widehat{D_+ D_-} \dot{P}^i \right)_k = q_k^i - \frac{1}{2K+1}$ when $k = -K, -K+1, \dots, K$. Indeed, in the interior, for $|k| < K$, the $\widehat{D_+ D_-}$ -operator is identical to the $D_+ D_-$ -operator, and the property follows immediately from the facts that $(D_+ D_- P^i)(k) = q_k^i$ in the interior, that $D_+ D_-$ maps linear functions $[ak + b]_k \in \mathbb{R}^{\mathbb{Z}}$ to 0 and the quadratic function $[k^2]_k \in \mathbb{R}^{\mathbb{Z}}$ to 2, just like the second-order differential operator satisfies $\frac{d^2}{dx^2}(ax + b) = 0$ and $\frac{d^2}{dx^2}x^2 = 2$ in the continuous setting.

At $k = -K$, we have

$$\begin{aligned} \left(\widehat{D_+ D_-} \dot{P}^i \right)_{-K} &= \dot{P}_K^i - 2\dot{P}_{-K}^i + \dot{P}_{-K+1}^i \\ &= \frac{1}{2K+1} P_K^i + \frac{K(2K+2)}{2K+1} - \frac{1}{2}(2K+2) \\ &\quad - 2 \left(P_{-K}^i - \frac{1}{2} \times \frac{2}{2K+1} \right) \\ &\quad + P_{-K+1}^i - \frac{1}{2K+1} \left(P_K^1 - \frac{1}{2}(2K+2) \right) - \frac{1}{2} \times \frac{2 \times 3}{2K+1} \\ &= q_{-K}^i - K - 1 \\ &\quad + \frac{1}{2K+1} (K(2K+2) + 2 + K - 3) \\ &= q_{-K}^i - \frac{1}{2K+1}. \end{aligned}$$

A similar argument implies that

$$\left(\widehat{D_+ D_-} \dot{P}^i \right)_K = q_K^i - \frac{1}{2K+1}.$$

It therefore follows that

$$\mathcal{F}(p)(\omega) = \begin{cases} 2(\cos(\omega) - 1) \times \frac{\hat{P}^1(\omega)^2}{\hat{P}^2(\omega)}, & \omega \neq 0, \\ 1 & \omega = 0. \end{cases}$$

Equivalently, in the physical domain

$$p_k = \begin{cases} \left(\widehat{D_+ D_-} F \right)_k + \frac{1}{2K+1}, & |k| \leq K, \\ 0, & |k| > K. \end{cases}$$

We are done. ■

Proof of Theorem 3: The result follows immediately from the fact that the appropriate R -function defined in (18) is constant, equal to one, if and only if $B^{\ell_1, \ell_2}(z) \equiv A(z)$, which only occurs if $(\ell_1, \ell_2) = (k_1, k_2)$. Otherwise, the function differs from one for almost all z on the unit circle in the complex plane, and E is consequently strictly positive. ■

D Examples

Bimodal event risk

Figure 12 shows the recovered event distribution and estimation errors for the example with bimodal event risk, in Section 2.5, using method (i) in Theorem 1. The top panel shows actual event distribution (line) and recovered distribution (dots). The middle and lower panels show the differences between recovered and true announcement probabilities for the two methods, method (i) in the middle panel and method (ii) in the lower panel. The pointwise errors are on the order of 10^{-11} , so the method works very well in this case.

CEV model with event risk

We also study a model that mimics constant elasticity of variance (CEV) dynamics of the underlying asset in normal periods. At time T , the asset price satisfies $v_T(k) = v_T(k-1) (1 + bv_T(k-1)^{\alpha-1})$, $0 < \alpha \leq 1$. Risk-neutral transition probabilities are given by $q_{-1} = q_0 = q_1 = \frac{1}{3}$. Prices at earlier points in time are derived from (29), and as before we normalize the risk-free rate to $R = 1$. Normal-period transitions are given by a trinomial tree, with one-period volatility $\hat{b} = b\sqrt{\frac{2}{3}}$, and the asset's return dynamics are then a discrete approximation of the continuous time CEV diffusion process

$$\frac{dv_t}{v_t} = \hat{b}v_t^\alpha d\omega_t,$$

where ω_t is a standardized Brownian motion. This is similar to how the standard binomial tree model provides a discrete approximation of the continuous time constant-coefficient Black-Scholes model.

We let each period represent half a day, set $T = 56$, corresponding to 28 days, and assume that the risk-neutral probabilities q^{28} and q^{56} are available. As before, the announcement is assumed to occur before $t = 28$, so method (ii) in Theorem 1 is applicable. We set the volatility coefficient to $\hat{b} = 0.015$, corresponding to an instantaneous annualized volatility of $\sigma = \hat{b} \times \sqrt{2 \times 365} \approx 0.4$ at the asset price $v = 1$. The elasticity parameter is $\alpha = 0.4$. The announcement occurs before $t = 28$, and we apply method (ii) in Theorem 1.

Figure 13 shows the state prices q^{28} and q^{56} . The effect of the announcement is quite pronounced in q^{28} , but much less so in q^{56} . In Figure 14, several volatilities are compared. The black line corresponds to the instantaneous volatility of the CEV model, which is decreasing and convex in the current spot price, and has the value $\hat{b} = 0.015$ when the spot price is 1. The red line corresponds to the implied volatility curve associated with the CEV model, as a function of the

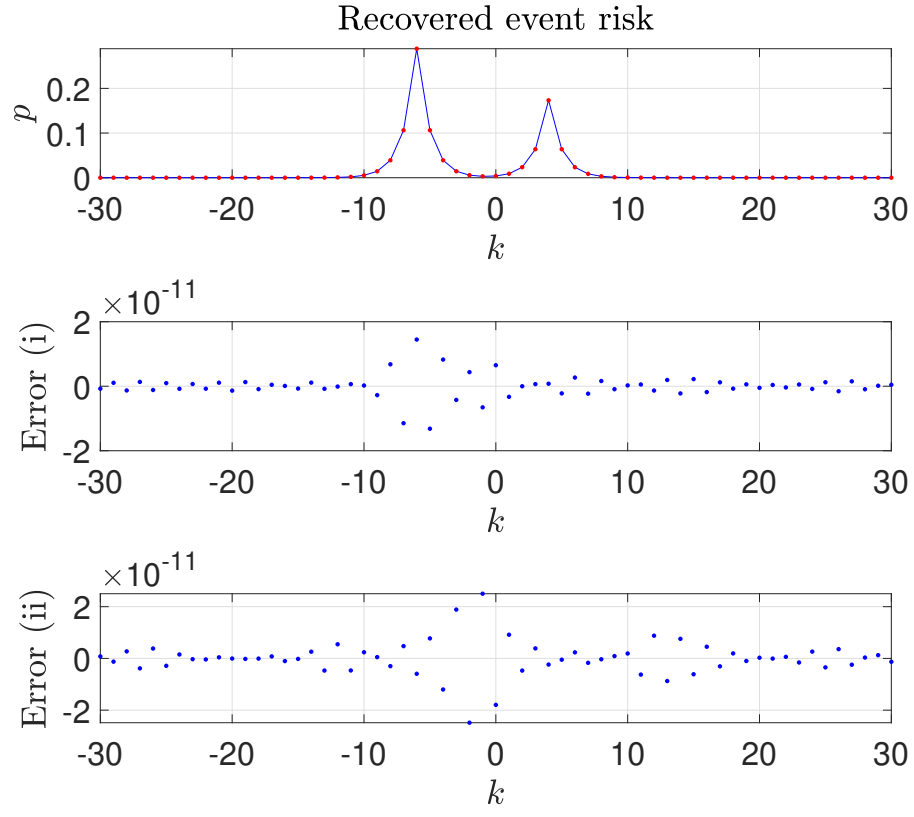


Figure 12: Top panel: Recovered event risk distribution. Middle and lower panels: Errors of the two methods of Theorem 1, method (i) in the middle panel and method (ii) in the lower panel.

option strike price, when there is no announcement risk. This curve is also convex and decreasing, but less so than the spot-price curve, because of the averaging effect, since the implied volatility is a function of the expected average volatility between 0 and T , rather than on the instantaneous volatility. The blue dotted line represents the implied volatilities associated with option prices in the CEV model with announcement risk. We see that it deviates significantly from the red line, due to the announcement risk: It (i) lies significantly above the red line, (ii) is partly increasing and partly decreasing, and (iii) has a “hat”-shape, with both concave and convex regions.

Finally, in Figure 15, we show actual and recovered event risk. The two curves are, again, indistinguishable. We stress, again, that the only inputs to the method are the state prices, q^{28} , and q^{56} . Although the CEV model and the event risk specifications are used to generate these state prices, the recovery method itself does not “know” this. It simply applies Theorem 1 to whatever state prices are given.

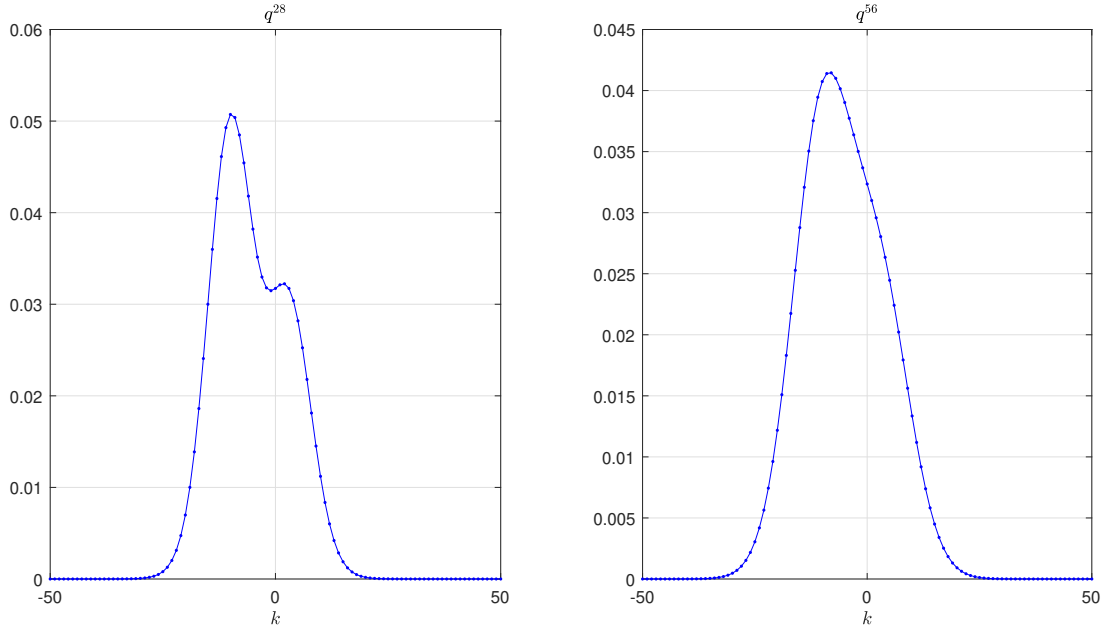


Figure 13: *Left Panel: 28 period (half days) state prices. Right Panel: 56 period (half days) state prices.*

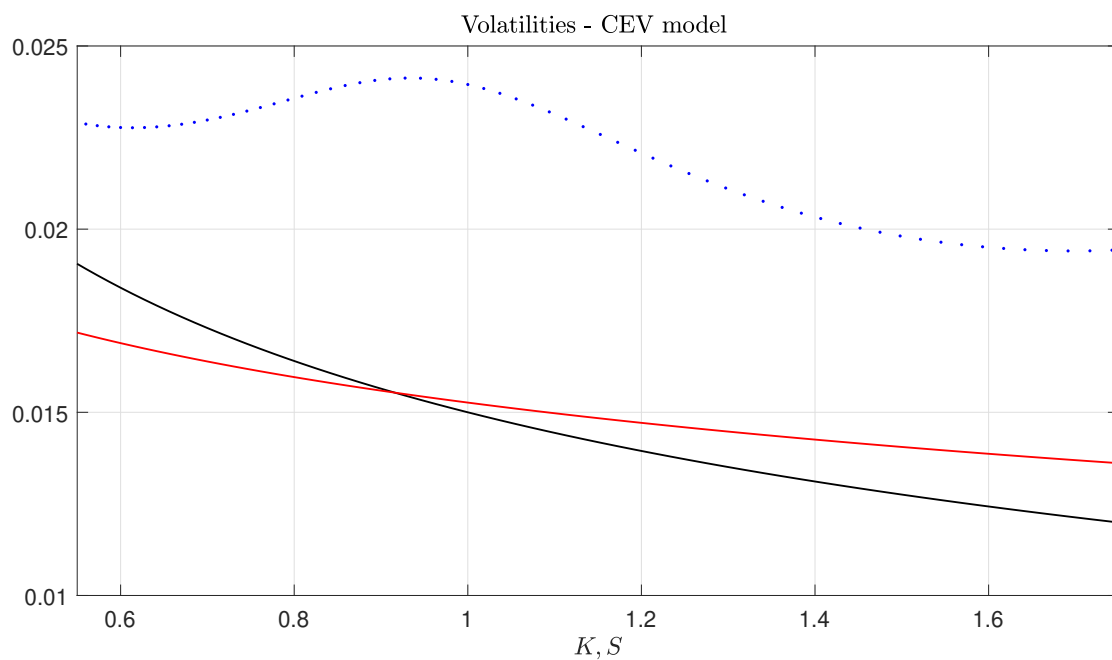


Figure 14: *Instantaneous volatility of CEV model (solid black line), implied volatilities of prices generated by CEV model (solid red line), and implied volatilities of model that includes both CEV risk and announcement risk (dotted blue line).*

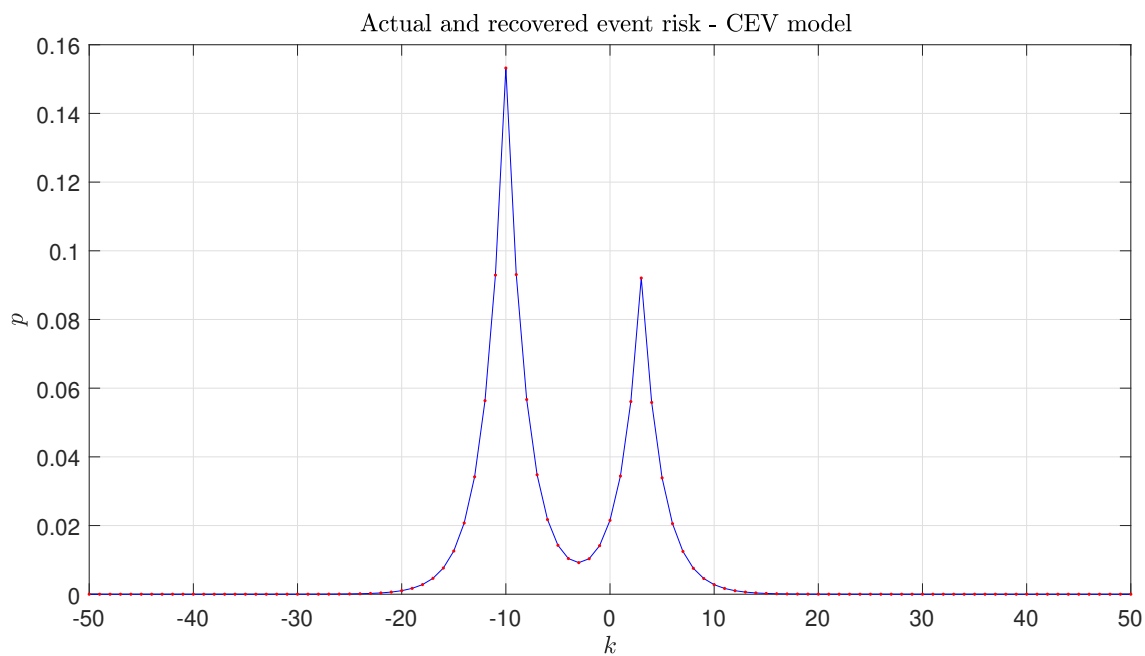


Figure 15: *Recovered event risk distribution, using method (ii) of Theorem 1.*

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